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Unified Multi-Agent Coordination Bridging Large Language Models and Symbolic AI for Autonomous Systems

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Resumen

Los modelos grandes de lenguaje pueden interpretar solicitudes y proponer planes, pero la plausibilidad lingüística no establece factibilidad, autoridad ni terminación verificable. Esta tesis propone un límite de autorización y rechazo para coordinar agentes heterogéneos: los componentes lingüísticos producen selecciones tipadas no autoritativas, mientras una capa determinista resuelve identidades, dependencias, restricciones, confianza y contratos antes del despacho. El prototipo integra admisión semántica, compilación simbólica acotada, contratos de artefactos, ejecución A2A, trazas persistentes y estado autoritativo replicado mediante etcd/Raft.

El experimento principal v0.7 se congeló antes de la recolección. Comprende 64 casos retenidos en 32 pares factible/no factible, dos brazos, tres modelos locales y dos semillas: 768 salidas. En la semilla primaria, el híbrido obtuvo 71.88% de exactitud balanceada, 44.79% de cobertura factible y una aceptación insegura; el LLM directo obtuvo 53.13%, 98.96% y 89 aceptaciones inseguras. La superioridad de seguridad y los invariantes simbólicos aprobaron; cuatro de seis criterios, incluidos utilidad, cero aceptaciones inseguras, no inferioridad de cobertura y superioridad sobre coincidencia exacta de alias, fallaron. El control exacto resolvió los 64 casos, lo cual limita fuertemente la validez de constructo. Qwen3-1.7B se usó primero para calificación de desarrollo; sus resultados no forman parte de la evidencia confirmatoria.

La campaña consensus-v4 completó 45 ensayos: 43 aprobaron, dos reconfiguraciones terminaron por tiempo de infraestructura y no hubo violaciones en 140 de 150 comprobaciones primarias ejecutadas. Su procedencia registra un árbol de trabajo sucio, por lo que se presenta como evidencia descriptiva completa y no como evidencia limpia aceptada. Las etiquetas son del autor y no hubo adjudicación independiente. No se afirma verdad semántica, tolerancia bizantina, efectos exactamente una vez, preparación para producción ni superioridad general.

El seguimiento v0.8 introduce evidencia textual verificable, ambigüedad derivada de forma determinista, límites para citas, negación y términos desconocidos, recuperación lexical no autoritativa y búsqueda de proveedores dirigida por restricciones. Su matriz de desarrollo de 576 observaciones con Qwen3-1.7B y Gemma-4-E2B aprobó todas las puertas de calificación; Qwen3-8B fue excluido. Estos resultados fueron usados para desarrollo y no se presentan como confirmatorios. La recolección retenida permanece cerrada hasta completar dos pasadas humanas de etiquetado separadas temporalmente.

Palabras clave: sistemas multiagente; modelos grandes de lenguaje; inteligencia artificial simbólica; autorización; trazabilidad; A2A.

Abstract

Large language models can interpret requests and propose plans, but linguistic plausibility does not establish feasibility, authority, or verifiable completion. This thesis proposes an authorization/refusal boundary for heterogeneous-agent coordination: linguistic components emit typed, non-authoritative selections, while a deterministic layer resolves identities, dependencies, constraints, trust, and contracts before dispatch. The prototype integrates semantic admission, bounded symbolic compilation, artifact contracts, A2A execution, durable traces, and etcd/Raft-replicated authoritative state.

The primary v0.7 experiment was frozen before collection. It contains 64 held-out cases in 32 feasible/infeasible pairs, two arms, three local models, and two seeds: 768 outputs. At the primary seed, the hybrid achieved 71.88% balanced accuracy, 44.79% feasible recall, and one unsafe acceptance; the direct LLM achieved 53.13%, 98.96%, and 89 unsafe acceptances. Safety superiority and symbolic invariants passed; four of six criteria failed, including utility, zero unsafe acceptance, recall non-inferiority, and superiority over exact-alias matching. The exact control solved all 64 cases, sharply limiting construct validity. Qwen3-1.7B was used first for developmental qualification; its results are excluded from confirmatory evidence.

The consensus-v4 campaign completed 45 trials: 43 passed, two reconfiguration trials ended in infrastructure timeouts, and no violation occurred in 140 of 150 executed primary checks. Its provenance records a dirty worktree, so it is reported as complete descriptive evidence rather than accepted clean evidence. Labels were author-produced without independent adjudication. The thesis makes no claim of semantic truth, Byzantine tolerance, exactly-once effects, production readiness, or general superiority.

The v0.8 follow-up adds verifiable textual evidence, deterministically derived ambiguity, quotation/negation/unknown-term boundaries, non-authoritative lexical retrieval, and constraint-directed provider search. Its 576-observation development matrix with Qwen3-1.7B and Gemma-4-E2B passed every qualification gate; Qwen3-8B was excluded. These tuned results are developmental, not confirmatory. Held-out collection remains sealed until two temporally separated human labeling passes are completed.

Keywords: multi-agent systems; large language models; symbolic artificial intelligence; authorization; traceability; A2A.

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Chapter 1

Introduction

1.1 Context and Motivation

Autonomous systems increasingly coordinate components that differ in ownership, implementation, authority, and reasoning method. Large language models (LLMs) extend these systems with natural-language interpretation, flexible task decomposition, and explanation. Symbolic artificial intelligence (AI), in contrast, provides explicit representations of capabilities, constraints, plans, and state transitions. The two approaches are complementary: language models are effective at proposing meaning under ambiguity, whereas symbolic methods are effective at checking conditions that can be stated precisely.

This complementarity does not automatically yield dependable coordination. An LLM can describe a plausible division of work even when no admitted agent possesses a required capability, two tasks exchange incompatible artifacts, or an action exceeds the authority granted to its executor. Free-form dialogue can also obscure whether an agent merely discussed a task, accepted a commitment, or supplied evidence of completion. These distinctions matter whenever coordinated agents can invoke tools, modify shared state, or affect an external environment.

1.2 Problem Statement and Research Gap

Classical multi-agent systems provide explicit interaction protocols, commitments, organizational roles, and formal coordination mechanisms. Recent LLM-agent systems provide broad linguistic flexibility and tool-use interfaces. Emerging interoperability protocols improve connectivity among independently implemented agents. Nevertheless, these lines of work do not by themselves define when a natural-language coordination proposal is sufficiently justified for execution. In particular, connectivity does not establish semantic compatibility, and fluent agreement does not establish feasibility.

The scientific problem addressed by this thesis is therefore:

How can a heterogeneous multi-agent system use LLM-based interpretation and decomposition while ensuring, through explicit symbolic criteria, that an accepted coordination plan is supported by available capabilities, compatible artifacts, applicable authority, and verifiable completion conditions?

The problem is not merely the construction of another agent platform. It is the absence of a precise operational boundary between a linguistically plausible proposal and a plan that the current system is justified in authorizing. Existing orchestration frameworks, agent protocols, schemas, validators, and tool registries provide important building blocks; the novelty sought here is the enforced combination of those pieces into a coordination boundary where candidate plans, authority checks, artifact contracts, commitments, refusals, and execution traces are all explicit before dispatch.

1.3 Object of Study and Field of Action

The *object of study* is coordination in heterogeneous autonomous multi-agent systems. The *field of action* is the hybrid mediation of natural-language requests into symbolically constrained plans, commitments, execution traces, and infeasibility decisions for independently implemented agents.

1.4 Research Objective

The general objective is to design and specify a unified coordination framework that uses LLMs for semantic interpretation and candidate decomposition while using symbolic representations and deterministic checks to authorize, trace, or reject multi-agent plans.

The specific objectives are:

1. Critically synthesize classical multi-agent coordination, symbolic AI, LLM-agent architectures, and interoperability protocols to identify the unresolved integration gap.
2. Define a protocol-independent theoretical framework for capabilities, tasks, authority, commitments, artifact compatibility, bounded auxiliary capabilities, and feasibility.
3. Instantiate the framework as an implementable Python coordination agent over middle-ware SDK that preserves protocol-mediated interoperability while supporting remote, local, and Lingo-authored agents without coupling the theoretical contribution to a particular model provider, agent framework, or communication protocol.

4. Evaluate the corrected prototype through two evidence tracks: a frozen production-path, author-labelled, label-hidden 64-case matched-pair study with three local models, two seeds, dual arms, and deterministic controls; and a consensus-backed 45-trial crash-fault campaign over three-, five-, and seven-voter coordinator topologies, complemented by deterministic, A2A, and PostgreSQL checks.
5. Analyze the scope, limitations, validity threats, and practical implications of the proposed framework.

1.5 Research Questions

The thesis is guided by the following questions:

- RQ1** Which coordination information must be represented explicitly to distinguish an executable multi-agent plan from a linguistically plausible one?
- RQ2** How should responsibility be divided between LLM-based reasoning and symbolic validation in a heterogeneous coordination workflow?
- RQ3** How can missing narrow linguistic capabilities be introduced without granting unbounded new competence or authority?
- RQ4** What do the measured outcome and process results show about justified acceptance, explicit refusal, traceability, dependency discipline, and recovery relative to paired controls and ablations?
- RQ5** Under which measured crash, partition, restart, and membership-reconfiguration conditions can physically replicated coordinators preserve one logical authorization state and suppress duplicate committed effects?

1.6 Scope and Contributions

The thesis focuses on plan-level coordination under one trusted logical authorization role and physically replicated execution state. It does not claim to verify the internal reasoning of participating agents, solve Byzantine coordination, or guarantee runtime success after authorization. Its contribution is integrative: a formal separation between linguistic proposal and symbolic authorization; typed, fail-closed feasibility and completion contracts; bounded task-local auxiliary capabilities; durable trace and measured crash recovery; a local A2A admission boundary; consensus-backed replication of the authoritative path; and reproducible conformance and fault-campaign methods. Established component mechanisms, the new operational integration, prototype evidence, and unproven properties are kept distinct.

1.7 Methodological Approach

The research follows a design-science approach. It combines critical literature synthesis, formal and architectural modeling, implementation-oriented specification, and controlled evaluation. The evaluation separates planning feasibility from runtime success, uses pre-specified author labels hidden during inference and repeated local-model runs, tests mechanism ablations, and measures replicated-coordinator ownership through a shared durable store. Chapter 3 explains the executed and gated stages.

1.8 Thesis Organization

Chapter 2 synthesizes the theoretical foundations and research gap. Chapter 3 defines the research and evaluation method. Chapter 4 presents the protocol-independent framework. Chapter 5 maps it to a coordination-agent architecture over SDK middleware and then selects a concrete interoperability adapter for implementation. Chapter 6 specifies the evaluation execution and reporting structure. Chapter 7 examines the implications, limitations, and validity threats. The final chapter relates the contributions to the objectives and identifies future work.

Chapter 2

State of the Art

2.1 Framing the Research Problem

The contemporary discussion of artificial intelligence agents is marked by a productive but sometimes imprecise convergence between two research traditions. The first tradition is the classical computer-science study of autonomous agents and multi-agent systems, in which an agent is a situated computational entity that perceives an environment, reasons over goals or policies, and acts through effectors [112, 90, 56]. The second tradition is the recent development of large language model (LLM) applications that use natural language, tool invocation, memory, and planning prompts to produce behavior that is described as agentic [15, 108, 115, 107]. The overlap between these traditions is substantial, but not identical. Classical agents emphasize autonomy, reactivity, proactiveness, social ability, commitments, and environment interaction; LLM agents often emphasize linguistic reasoning, flexible task decomposition, and interaction with external tools.

This distinction is central to a thesis on unified multi-agent coordination. A chatbot that produces a response to a prompt is not, by itself, an autonomous agent in the classical sense. Conversely, a symbolic controller that follows a plan can be an agent even if it has no natural-language interface. The research opportunity appears when LLMs are treated as components inside broader agent architectures rather than as replacements for agency itself. In that view, an LLM can serve as a linguistic reasoning module, a translator between human instructions and formal actions, an explanation generator, or a coordinator of high-level intent. The agent, however, remains the larger computational system that maintains state, chooses actions, interacts with other agents, and is accountable for execution.

The literature therefore motivates a bridging problem. Classical multi-agent systems (MAS) provide mature formalisms for interaction protocols, negotiation, task allocation, organizational roles, commitments, and verification [111, 109, 29, 98]. LLM-based systems provide flexible natural-language mediation, few-shot generalization, and tool-use interfaces [83, 92, 72]. Symbolic AI supplies explicit representations for planning, constraint

satisfaction, rule-based control, and reasoning over dynamic worlds [41, 89, 49]. A unified coordination architecture must integrate these strengths without collapsing their differences. In particular, it must support agents that communicate linguistically, reason symbolically, invoke tools safely, and participate in multi-agent coordination mechanisms whose semantics are explicit enough to be analyzed.

The phrase state of the art is used here in a precise sense. It does not refer only to the most recent LLM frameworks or protocols, because novelty alone is not sufficient for autonomous coordination. Instead, the relevant state of the art consists of the strongest available concepts, formalisms, systems, and evaluation practices that inform the design of a coordinated autonomous system. This includes older but still foundational work on agency and communication, mature symbolic methods for action and verification, and newer LLM-based systems that demonstrate flexible linguistic control. The purpose of the chapter is therefore synthetic: it identifies what each research line contributes, where each line remains insufficient, and why a unified LLM-symbolic MAS architecture is a coherent research target.

2.2 Classical Foundations of Autonomous Agents

The classical agent literature provides the conceptual baseline for this thesis. Wooldridge and Jennings characterize agents through autonomy, social ability, reactivity, and proactiveness, thereby distinguishing agents from passive objects or services [112]. Russell and Norvig formulate the rational-agent view, in which an agent maps percept histories to actions and is evaluated by a performance measure in an environment [90]. These definitions remain important because they locate agency in a control loop: perception, internal state update, decision, and action. The same loop is required for autonomous systems whether their reasoning module is symbolic, neural, or hybrid.

Several architectural families have shaped the design of autonomous agents. Reactive agents connect percepts to actions through condition-action rules or layered behaviors. Brooks’s subsumption architecture demonstrated that robust situated behavior could arise from distributed reactive layers without a centralized symbolic world model [14]. Deliberative agents, by contrast, maintain explicit models and use search or planning to choose actions. Hybrid architectures combine these approaches, usually by placing a fast reactive layer under a slower deliberative layer. Cognitive architectures such as Belief-Desire-Intention (BDI) systems add explicit mental attitudes, representing what the agent believes, wants, and has committed to doing [13, 87, 88, 39].

BDI architectures are particularly relevant to the present thesis because they provide a disciplined vocabulary for autonomous behavior. Beliefs encode information about the world, desires encode possible objectives, and intentions encode commitments to plans. This separation is useful for LLM-symbolic systems because LLM output should not automatically

become a commitment to act. A linguistic proposal may be treated as a candidate desire, an explanation, or a plan sketch; the symbolic or procedural layer can then validate whether it should become an intention. This division helps address a common weakness of LLM agents: fluent text can appear decisive even when the underlying action has not been checked against state, preconditions, or policy.

Agent-oriented programming also contributes important design principles. Shoham proposed programming in terms of mentalistic concepts such as beliefs, obligations, and capabilities [96]. Later platforms such as JADE, Jason, GOAL, and JaCaMo operationalized agent communication, BDI reasoning, and multi-agent organization in software frameworks [8, 12, 47, 10]. These systems show that agent design benefits from explicit abstractions for messages, roles, plans, and norms. Modern LLM frameworks often rediscover these abstractions through role prompts, tool registries, memory modules, and conversation controllers, but the classical literature provides more precise semantics for them.

The boundary between agent and non-agent software is therefore not merely terminological. A database, an API, a workflow, and a model endpoint can all participate in an agentic system, but they do not necessarily exhibit agency. An agent must have a decision procedure and some degree of autonomy with respect to its environment [36]. This matters for system design because different components require different forms of assurance. A tool can be validated by its interface contract; a model can be evaluated by prediction or generation quality; an autonomous agent must also be evaluated by its policy for choosing when and how to act. In LLM-based systems, this distinction is often obscured when every tool wrapper or prompt template is called an agent. A rigorous thesis architecture should reserve the term agent for components that participate in a sense-reason-act loop with explicit state and responsibility.

Table 2.1: Agent architectural traditions relevant to unified coordination.

Tradition	Relevance to this thesis
Reactive agents	Provide fast environment response and motivate safety shields that can override slower deliberation.
Deliberative agents	Provide explicit planning, goal reasoning, and action selection over symbolic state.
BDI agents	Separate beliefs, desires, and intentions, which is useful for interpreting LLM proposals before execution.
Hybrid agents	Combine reactive control with deliberative reasoning, matching the need to combine LLM reasoning with symbolic execution.
Agent-oriented programming	Provides software abstractions for roles, commitments, messages, plans, and organizational structure.

2.3 Multi-Agent Coordination

Multi-agent systems extend the agent concept from individual autonomy to distributed computation among multiple autonomous entities. In a MAS, agents may cooperate, compete, negotiate, delegate, or coordinate around shared resources [29, 109, 111]. The key difficulty is that no single agent necessarily possesses complete knowledge, global control, or unlimited authority. Coordination is therefore required to manage dependencies among activities, allocate tasks, resolve conflicts, synchronize plans, and maintain coherent collective behavior. Malone and Crowston define coordination broadly as the management of dependencies between activities [69]; the MAS literature operationalizes this idea through protocols, organizational structures, distributed algorithms, and shared environments [55].

Task allocation is one of the oldest and most concrete coordination problems. The Contract Net Protocol introduced a manager-contractor pattern in which tasks are announced, agents bid, and contracts are awarded according to suitability or cost [99]. Market-based coordination generalizes this approach by using auctions, marginal costs, and economic mechanisms to allocate work or resources [91, 110]. Coalition formation addresses cases in which a task requires the combined capabilities of multiple agents [94]. These mechanisms are directly relevant to LLM-based multi-agent systems, where role-specialized agents may be dynamically selected for subtasks such as planning, coding, verification, retrieval, or simulation.

Shared-state coordination offers a different model. Blackboard systems allow specialized knowledge sources to read and write partial solutions in a common workspace [80, 19]. Tuple-space coordination, exemplified by Linda, decouples agents through generative communication over shared tuples [37]. Stigmergic coordination uses changes in the environment as indirect signals, a concept developed in studies of social insects and later applied to swarm systems [101]. These mechanisms anticipate modern agent workspaces, shared memories, vector stores, and artifact repositories. They also expose a risk: if the shared state is ambiguous or corrupted, coordination can fail even when each individual agent appears locally competent.

Distributed constraint reasoning provides a more formal approach to joint decision-making. Distributed Constraint Satisfaction Problems and Distributed Constraint Optimization Problems model decisions as variables controlled by different agents under local and global constraints [116, 77, 32]. Such formalisms are valuable for autonomous systems because they make resource limits, task dependencies, and optimization criteria explicit. LLM systems rarely use DCOP-style models directly, but a unified architecture can use LLMs to elicit goals or preferences while a symbolic optimizer enforces constraints. This division is especially important when natural-language instructions are incomplete, inconsistent, or underspecified.

Social and normative approaches broaden coordination beyond efficiency. Commitments, norms, institutions, trust, and argumentation provide mechanisms for regulating interactions among autonomous agents [98, 24, 86]. In classical MAS, a commitment can represent an obligation from one agent to another; protocols can then be evaluated in terms of how commitments are created, discharged, or violated. This perspective is relevant to LLM agents because dialogue alone does not guarantee accountability. If one agent asks another to perform an action, a unified system should represent whether that request created an obligation, whether the target accepted it, and what evidence establishes completion.

Coordination failure is as important as coordination success. Distributed systems may fail through deadlock, livelock, inconsistent local beliefs, resource contention, duplicated effort, or incompatible goals. LLM-based MAS add further failure modes: agents may agree linguistically while using different internal interpretations; a critic agent may approve an artifact without checking it; multiple agents may reinforce the same hallucination; or a delegated task may be considered complete because the conversation ended rather than because a verifiable postcondition was reached. Classical MAS mechanisms provide vocabulary for diagnosing these failures. For example, a protocol trace can reveal that an expected acceptance message was never sent, and a commitment record can reveal that a promised action lacks evidence of completion. This is a strong argument for storing coordination state in explicit symbolic structures rather than only in chat history.

The distinction between centralized and decentralized coordination also requires careful treatment. Centralized orchestration simplifies global control and is common in practical LLM systems, where a controller routes messages among role-specific agents. Decentralized coordination can improve robustness and scalability, but it requires stronger assumptions about local decision-making and communication semantics. Hybrid coordination is often more realistic: a central orchestrator may allocate tasks, while agents negotiate details, update shared artifacts, and call local tools. For autonomous systems, the correct choice is domain-dependent. Safety-critical execution may require centralized authorization, while exploratory planning may benefit from decentralized proposal generation and critique.

2.4 Agent Communication Languages and Interaction Protocols

Coordination depends on communication, but not all communication has the same semantics. Classical Agent Communication Languages (ACLs) were developed to provide structured, interoperable message exchange among agents. Their theoretical basis lies in speech-act theory, where utterances are understood as actions that can request, inform, promise, or commit [6, 93, 17]. KQML introduced performatives such as ask, tell, and reply, together with

message fields for content, language, and ontology [31, 60]. FIPA-ACL later standardized communicative acts and interaction protocols with more explicit semantics grounded in mental attitudes [34].

The importance of ACLs for this thesis is not that modern systems should necessarily reuse KQML or FIPA unchanged. Rather, the key lesson is that coordination messages require interpretable force. A statement such as "the route is clear" differs from "please clear the route" or "I commit to clearing the route." In natural language, such distinctions are often implicit. LLM agents can generate and interpret nuanced language, but they can also blur assertion, request, plan, and commitment. A unified coordination architecture should therefore map natural-language utterances to structured communicative acts when those utterances affect execution.

Interaction protocols provide the temporal structure of communication. Contract net, request-response, subscription, negotiation, and auction protocols specify allowed message sequences and expected state transitions [34, 57]. In classical MAS, this structure supports interoperability and verification. In LLM-based systems, conversation often proceeds through ad hoc chat transcripts. Such transcripts are flexible, but flexibility can make it difficult to determine whether a task has been accepted, delegated, blocked, or completed. A hybrid architecture can preserve natural-language flexibility for explanation and negotiation while retaining structured protocol state for execution.

Content languages and ontologies also remain central. Gruber defines an ontology as an explicit specification of a conceptualization [44]; description logics provide formal foundations for class, relation, and constraint reasoning [7]. Without shared semantics, agents may exchange syntactically valid messages that have different meanings to different participants. LLMs can help mediate heterogeneous vocabularies, but they cannot by themselves guarantee semantic alignment. For autonomous systems, especially those controlling tools or physical processes, semantic grounding must be explicit enough to support validation.

The transition from ACLs to natural-language coordination should therefore be understood as an expansion of expressive capacity rather than a replacement for structure. Natural language is valuable because it can express uncertainty, context, explanation, and negotiation without requiring all possibilities to be pre-enumerated. However, execution requires discrete commitments. An LLM may write, "I can handle the scheduling task," but a coordination layer must determine whether this is an assertion of capability, an offer, an acceptance, or a mere conversational signal. Classical ACLs solve this by using performatives; modern LLM systems can approximate the same effect by extracting structured message types, confidence, referenced artifacts, intended recipients, and expected follow-up actions. The extraction need not eliminate natural language; it should accompany it with machine-checkable coordination metadata.

Another issue is observability. In a free-form conversation, the state of the protocol is

often implicit in the transcript. A human reader may infer that a task has moved from proposal to execution, but a verifier cannot safely depend on such inference. Structured communicative acts make protocol state observable. For example, a request can be represented with a sender, receiver, content, deadline, acceptance status, and completion criterion. This explicit state is compatible with LLM-generated messages, but it requires an architectural layer that treats dialogue as input to a coordination model rather than as the coordination model itself.

2.5 Symbolic Foundations

Symbolic AI provides the formal machinery needed to represent knowledge, define actions, and reason about consequences. Early work on logic and common-sense reasoning established the idea that intelligent behavior can be supported by symbolic descriptions of the world and action [70, 79]. Reiter’s work on knowledge in action developed logical foundations for specifying dynamic systems [89]. Knowledge graphs and ontologies extend this tradition by supporting semantic integration across heterogeneous data sources [49]. In a unified agent architecture, these symbolic representations can serve as the stable substrate against which LLM-generated interpretations are checked.

Automated planning is especially relevant because it provides a disciplined account of goal-directed action. STRIPS introduced actions with preconditions and effects [30]; PDDL later provided a standard planning-domain language that supported interoperability among planners [71, 35]. Planning systems search for action sequences that transform an initial state into a goal state under domain constraints [41]. This formalism directly addresses a weakness of LLM planning: language models can produce plausible plans without verifying action applicability, resource limits, or causal dependencies. A planner can reject or repair such plans when the formal model is available.

Hierarchical Task Network (HTN) planning adds a decomposition perspective. Rather than planning only through primitive actions, HTN methods refine abstract tasks into subtasks according to domain-specific methods [26, 78, 40]. This is closely aligned with how LLM agents often decompose natural-language tasks into steps. The difference is that HTN decomposition is constrained by explicit methods, whereas LLM decomposition is often implicit in generated text. Combining the two can allow an LLM to propose a decomposition while a symbolic layer verifies that each refinement is permitted by the domain model.

Rule-based and non-monotonic reasoning also contribute to agent control. Logic programming and Answer Set Programming support declarative encodings of constraints, defaults, and alternative stable models [38, 66]. These approaches are useful when coordination requires reasoning about exceptions, incomplete knowledge, or mutually exclusive choices. Finite-state machines, behavior trees, and Goal-Oriented Action Planning provide additional symbolic control structures that are widely used in robotics and games [18, 82]. Their

advantage is not linguistic flexibility but predictable execution structure. In safety-critical autonomous systems, predictability is not optional.

Symbolic methods also clarify the difference between a plan and an explanation. LLMs can explain a possible course of action in coherent prose, but a plan in the symbolic sense is an object with preconditions, effects, ordering constraints, resources, and success criteria. Treating explanations as plans can lead to execution errors because the text may omit necessary conditions or represent them ambiguously. Conversely, symbolic plans can be difficult for humans to understand without explanation. A hybrid system can use a symbolic planner to produce or verify the executable structure while using an LLM to explain the plan, ask for missing information, or translate user intent into formal goals. The two artifacts should remain linked but distinct.

There are also limits to symbolic methods. A planner can only reason over what has been modeled. A rule engine can only enforce rules that are encoded. A knowledge graph may be incomplete or stale. The role of the LLM is therefore not only convenience; it can help interpret underspecified tasks, retrieve relevant context, and generate candidate formalizations. The strongest hybrid design is not a symbolic system with a superficial language interface, nor an LLM system with occasional tool calls. It is an architecture that assigns responsibilities according to epistemic strength: language models handle interpretation and communication; symbolic components handle explicit constraints, commitments, and execution semantics.

2.6 Neuro-Symbolic AI and Hybrid Reasoning

Neuro-symbolic AI studies the integration of neural learning with symbolic reasoning. Surveys emphasize that neural systems are effective for perception, pattern recognition, and language, while symbolic systems support abstraction, compositionality, explanation, and formal reasoning [9, 22, 48]. LLMs intensify this integration problem. They are neural models with broad linguistic competence, but their outputs are not inherently grounded in formal semantics. A hybrid architecture can exploit their strengths while delegating constraint satisfaction, planning, and verification to symbolic components.

One important pattern is LLM-to-symbolic translation. A language model may translate a user request into a PDDL goal, a program, a query, a constraint set, or an action schema. LLM+P, for example, explores the use of LLMs with classical planning by translating natural-language planning tasks into planner-compatible representations [67]. Studies of LLM planning capabilities also show that LLMs are not reliable substitutes for planners when exact action semantics are required [103]. The implication is not that LLMs are unhelpful for planning; rather, they should be placed where linguistic flexibility is valuable and checked where formal correctness matters.

A second pattern is symbolic-to-LLM augmentation. Retrieval-augmented generation

supplies external documents as context [62]; tool-augmented systems expose calculators, search engines, databases, or APIs [92, 72]; MRKL systems propose modular architectures in which language models route tasks to specialized symbolic or computational modules [59]. In agent systems, this pattern generalizes to planners, solvers, knowledge graphs, monitors, and execution kernels. The LLM becomes an interface and orchestrator, not the sole reasoner.

The most relevant pattern for this thesis is a hybrid execution loop. The agent perceives or receives a task, uses an LLM to interpret context and propose candidate actions, translates selected candidates into formal representations, uses symbolic modules to validate or refine them, executes through a controlled interface, and records feedback for future reasoning. This loop aligns with ReAct-style interleaving of reasoning and action [115], but adds a stronger separation between linguistic proposal and executable commitment. Such separation is essential in multi-agent settings because one agent’s text may trigger another agent’s action. The bridge between language and symbolic execution must therefore include validation, provenance, and state updates.

2.6.1 Closest Verification-First Planning Work

Formal-LLM supervises LLM plan generation with a developer-supplied automaton, directly addressing invalid plan sequences [65]. Metagent-P combines LLM world knowledge, symbolic cognitive architecture, and metacognitive verification for open-world planning [120]. Verified Multi-Agent Orchestration uses a plan–execute–verify–replan loop and an LLM-based completeness verifier over a question DAG [117]. The formal-methods position of Zhang et al. argues more broadly that trustworthy agents require integration between LLMs and formal methods [118]. These works make a generic “neuro-symbolic” novelty claim untenable; the remaining gap is narrower and operational.

Recent neighboring systems narrow that gap further. LOOP iteratively exchanges PDDL specifications and symbolic feedback rather than performing one-shot translation [105]. Cornelio et al. combine hierarchical decomposition, knowledge-graph RAG, and a symbolic validator that also detects execution failure [20]. Chimera combines an LLM strategist, symbolic constraints, and causal inference for multi-objective decision making [4]. The distinction claimed here is therefore not neuro-symbolic planning in general. It is the joint treatment of heterogeneous agent admission, canonical authority, artifact contracts, refusal before dispatch, receiver-observed operation fencing, and consensus replication of the authorization state.

The bridge is not a single translator. It is better understood as a set of transformations and checks. Natural-language goals may be transformed into symbolic goals; symbolic plans may be transformed into natural-language explanations; tool schemas may be transformed into LLM-callable functions; monitor violations may be transformed into feedback for

Table 2.2: Closest verification-first work and the remaining thesis distinction.

Work	Primary verification mechanism	Distinction of this thesis
Formal-LLM	Automaton-constrained plan generation	Authorizes composition against a live admitted registry, local authority, artifact contracts, and durable attempt evidence.
Metagent-P	Cognitive-symbolic planning plus metacognitive verification	Focuses on heterogeneous executor admission, refusal, protocol adaptation, and dependency-safe execution rather than one agent’s open-world planning loop.
Verified Multi-Agent Orchestration	LLM verification of completeness followed by replanning	Uses deterministic plan-level gates for modeled predicates and distinguishes contract validity from LLM quality judgment.
Trustworthy-agent formal-methods position	Research agenda joining LLMs and formal methods	Supplies one concrete implementation refinement, crash-recovery boundary, and reproducibility protocol; it does not claim comprehensive formal verification.

replanning. Each transformation can introduce error. Consequently, a robust bridge should record provenance, expose confidence or uncertainty when available, and support rejection or repair. This position differs from treating the LLM as a universal planner. It treats the LLM as one participant in a heterogeneous reasoning pipeline.

Hybrid reasoning also raises a granularity question. The LLM may operate at the level of high-level tasks, while a symbolic planner operates at the level of primitive actions. If the boundary is too high, the symbolic layer cannot catch relevant failures. If the boundary is too low, the LLM may be forced to produce brittle formal details it is not well suited to generate. A practical architecture should allow multiple levels of representation: strategic goals, decomposed tasks, protocol messages, symbolic constraints, and executable tool calls. Coordination then becomes the process of maintaining consistency across these levels.

2.7 LLMs as Agent Components

The rise of transformer-based language models created new possibilities for agent design. The transformer architecture enabled scalable sequence modeling through attention mechanisms [104]. Large pre-trained models demonstrated few-shot and instruction-following behavior [15, 83]. Chain-of-thought prompting showed that intermediate reasoning traces can improve performance on some reasoning tasks [108]. These developments made it practical to use LLMs as flexible reasoning and communication modules inside larger systems.

ReAct is an influential pattern because it interleaves reasoning traces with actions over external tools or environments [115]. Toolformer and related work further demonstrate that language models can be adapted to call external tools [92]. Augmented language model surveys describe this broader movement toward coupling LLMs with retrieval, tools, memory, and external computation [72]. For autonomous agents, tool use changes the risk profile. A generated sentence may be harmless; a generated tool call can modify files, execute code, send messages, or change a physical state. Therefore, the agent architecture must treat tool invocation as an execution event governed by permissions, validation, and monitoring.

Memory is another key capability. Retrieval-augmented generation provides access to external knowledge [62]; Reflexion introduces verbal feedback and episodic memory to improve future behavior [95]; Voyager stores reusable skills learned in a Minecraft environment [106]. These approaches show that agent performance depends not only on the base model but also on persistent context and learned artifacts. In a unified architecture, memory should be divided into forms with different semantics: conversational memory, factual knowledge, symbolic state, plan history, tool traces, and commitments. Collapsing all memory into unstructured text increases flexibility but reduces verifiability.

Embodied and interactive agents further illustrate both the promise and limits of LLM-based agency. Work on language models as zero-shot planners and embodied reasoning explores how LLMs can produce action sequences for simulated or physical tasks [52, 53]. Voyager demonstrates open-ended exploration with a skill library [106]. Generative Agents simulate social behavior through memory, reflection, and planning [84]. These systems show that LLMs can support long-horizon behavior, but they also depend on scaffolding: memory retrieval, environment feedback, tool execution, and action validation. The agent is the full scaffolded system, not only the model.

The current LLM-agent literature also reveals a tension between generality and reliability. Generality comes from representing many tasks in language and relying on the model’s broad prior knowledge. Reliability usually comes from narrowing the action space, adding tests, constraining tools, and validating intermediate artifacts. Autonomous systems need both. A useful agent may have to interpret open-ended human instructions, but it should not execute open-ended actions without controls. This tension explains why tool schemas, sandboxes, structured outputs, retrieval systems, and evaluators have become common in agent frameworks. They are attempts to recover reliability around a flexible linguistic core.

For multi-agent coordination, another limitation is social reasoning. LLMs can simulate dialogue and infer intentions, but simulated agreement does not necessarily correspond to coordinated execution. A planning agent may believe a worker agent has accepted a task because the worker responded positively, while the worker may lack the capability or authority to execute it. Classical MAS concepts such as capabilities, commitments, and organizational roles can reduce this ambiguity. The LLM can make role-based interaction

more natural, but the system should still maintain explicit records of who can do what, who has promised what, and what evidence resolves the promise.

2.8 LLM-Based Multi-Agent Systems

LLM-based multi-agent systems adapt MAS ideas to teams of language-model-driven components. Surveys of LLM agents and LLM-based MAS identify common patterns: role specialization, debate, delegation, reflection, tool use, and conversation-based collaboration [107, 46, 102]. These systems often assign roles such as planner, critic, coder, reviewer, researcher, or executor. The roles are usually encoded in prompts rather than in formal organizational models, but the functional resemblance to classical MAS is clear.

AutoGen introduced a flexible framework for composing conversable agents that may use LLMs, tools, or humans in the loop [113]. CAMEL uses role-playing prompts to study communicative cooperation among agents [64]. MetaGPT encodes software-engineering workflows as standard operating procedures distributed across specialized agents [50]. Chat-Dev similarly frames software development as communication among role-specialized agents [85]. Current engineering frameworks broaden this pattern: LangGraph provides low-level orchestration for durable, stateful agents [61]; Semantic Kernel provides middleware for integrating AI agents, models, plugins, and application code [74]; CrewAI organizes autonomous agents into crews and flows for complex workflows [21]. These systems demonstrate that multi-agent decomposition can improve task structure, distribute responsibilities, support persistence, and enable iterative critique. However, they do not by themselves settle whether a natural-language plan is authorized by an admitted registry, whether artifact contracts are semantically sufficient, or whether a refusal should be returned before dispatch.

Lightweight conversational-modeling frameworks address a narrower but important part of this landscape. Lingo is a type-safe Python framework for building LLM-powered applications around explicit conversational state, structured generation, structured flows, finite-state control, reflexive guardrails, and developer-controlled tool loops [45]. Its relevance is not that it replaces a MAS platform, an interoperability protocol, or a symbolic planner. Rather, it shows a state-of-the-art pattern for disciplining the linguistic layer itself: free-form language is wrapped in typed context, constrained model decisions, schema-validated outputs, and state-conditioned dialogue. This makes Lingo a useful example of how linguistic agents can be engineered inside a larger autonomous system while still requiring external authorization and feasibility validation before generated proposals become executable commitments.

This framework landscape clarifies the difference between a linguistic-agent construction toolkit and a coordination architecture. The former can narrow model behavior into typed acts, support state-conditioned dialogue, and expose guardrail hooks; the latter must still decide which agents may act, whether a plan is feasible, how artifacts satisfy contracts, and

Table 2.3: Representative LLM-based multi-agent approaches.

Approach	Coordination pattern	Limitation motivating hybrid design
AutoGen	Programmable conversations among agents, tools, and humans.	Conversation flexibility does not by itself provide formal protocol semantics.
CAMEL	Role-playing agents cooperate through prompted dialogue.	Role behavior is prompt-dependent and difficult to verify.
MetaGPT	Specialized agents exchange structured artifacts within a workflow.	Standard operating procedures require domain design and may remain task-specific.
ChatDev	Software development is decomposed into communicative roles.	Success depends on artifact consistency and evaluation of generated code.
LangGraph, Semantic Kernel, CrewAI	Practical orchestration, middleware, crews, flows, tools, and durable state for LLM applications.	Provide useful engineering substrates, but symbolic feasibility, authorization ownership, and infeasibility semantics remain application-level responsibilities.
Lingo	Typed conversational modeling with context, structured generation, flows, state machines, and reflexive guardrails.	Provides a disciplined linguistic layer, but not distributed MAS protocol semantics or symbolic feasibility checks.

what communication trace establishes completion. For this thesis, the SOTA implication is that language-mediated flexibility should be retained, but it should be placed behind explicit contracts, protocol state, and symbolic authorization.

The main lesson from this literature is that LLM-MAS research has reintroduced coordination as a practical necessity. When multiple LLM agents interact, the system needs role assignment, message routing, shared memory, conflict resolution, and evaluation of partial results. These are classical MAS problems. What is new is the linguistic richness of the agents and their ability to transform vague human instructions into candidate artifacts. The challenge is to connect that richness with formal models of interaction, state, and execution. Without such grounding, multi-agent dialogue may produce impressive transcripts but weak guarantees.

A second lesson is that role prompting is not the same as role semantics. In many LLM-MAS systems, an agent is assigned a role through natural-language instructions such as "act as a software architect" or "act as a reviewer." This can be effective for shaping model behavior, but it does not define a role in the organizational sense. A formal role has permissions, responsibilities, expected messages, authority boundaries, and accountability conditions. The gap between prompted roles and formal roles is significant for autonomous systems. A reviewer prompt may encourage criticism, but a reviewer role in a verified workflow should specify what evidence must be checked before approval.

The literature also suggests that multi-agent decomposition can sometimes improve output quality while also increasing system complexity. More agents mean more messages, more state transitions, more opportunities for inconsistency, and higher computational cost. The benefit of additional agents should therefore be justified by functional separation: one agent may specialize in natural-language interpretation, another in symbolic planning, another in verification, and another in execution. This division aligns with heterogeneous MAS more than with homogeneous debate among multiple copies of the same model. The thesis problem is thus not merely how to make many LLMs talk, but how to coordinate different kinds of reasoning entities.

2.9 Modern Agent Interoperability Protocols

The recent growth of agent frameworks has produced a new interoperability problem. Agents built in different frameworks may need to access tools, exchange context, advertise capabilities, delegate tasks, or collaborate across organizational boundaries. Several emerging protocols address parts of this problem. As of July 2026, the Model Context Protocol (MCP) defines a standardized client-server boundary for tools, resources, prompts, roots, sampling, elicitation, transports, and authorization; the thesis treats the 2025-11-25 specification as the stable cited baseline and the current draft page as protocol-evolution evidence rather than as

a settled production guarantee [5, 75, 76]. Agent2Agent (A2A) focuses on communication and interoperability between opaque agentic applications, including Agent Card discovery, JSON-RPC over HTTP(S), streaming, push notification, and structured data exchange; it is an open-source project under Linux Foundation governance, contributed by Google [42, 1, 100]. The Agent Communication Protocol (ACP) was introduced as an HTTP-native agent communication standard, but its documentation and LF AI and Data describe ACP as part of A2A, with independent active development winding down [54, 2, 63]. The Agent Network Protocol (ANP) emphasizes decentralized identity, discovery, messaging, and layered agent-network interoperability [3, 16].

Table 2.4: Emerging interoperability protocols for agent systems.

Protocol	Primary focus	Relevance to unified coordination
MCP	Tool and context integration between AI applications and external systems.	Provides a standardized tool/context boundary, but does not by itself define full agent-to-agent social semantics.
A2A	Agent-to-agent communication, capability discovery, and collaboration.	Addresses cross-framework agent interaction and can support higher-level coordination protocols.
ACP	Lightweight HTTP-native agent communication, now aligned with A2A efforts.	Illustrates convergence toward shared messaging standards for heterogeneous agents.
ANP	Decentralized identity, discovery, and networked agent communication.	Highlights identity and discovery as first-class concerns in open agent ecosystems.

These protocols should be interpreted carefully. They are important engineering developments, but they are not complete replacements for classical ACL semantics [25]. MCP can expose tools to an LLM application, but it does not determine whether a tool call is a request, a delegation, a commitment, or a verified action. A2A can support agent discovery and communication, but application-level coordination still requires domain protocols and state models. ACP’s convergence into A2A also shows that the protocol landscape is consolidating, not that the semantic coordination problem has disappeared. The relevant research question is therefore how modern protocol infrastructure can be mapped onto classical MAS concepts such as performatives, commitments, roles, and interaction protocols. A thesis architecture that bridges LLMs and symbolic AI should use modern protocols for interoperability while

preserving explicit semantics for coordination and execution.

The protocol landscape also illustrates the difference between connectivity and coordination. Connectivity means that components can exchange messages or expose tools. Coordination means that those exchanges contribute to a coherent joint process with interpretable state and controlled effects. MCP improves the tool boundary by making external context and tools available through a common interface. A2A and related efforts address cross-agent messaging and capability discovery. These are necessary infrastructure layers, but a MAS still requires policies for when to delegate, how to evaluate capability claims, how to recover from failure, and how to resolve conflicting commitments. In this sense, modern protocols lower the cost of integration but do not remove the need for agent theory.

Protocol evolution is also a governance issue. If agents are expected to operate across organizations, they must advertise identity, capabilities, permissions, and trust assumptions. Decentralized discovery may support open ecosystems, but it also creates risks around impersonation, malicious agents, and incompatible policy regimes. A unified architecture should therefore separate protocol transport from trust decisions. It should be possible to discover an agent without automatically trusting it, to receive a request without accepting it, and to expose a tool without granting unrestricted authority. These distinctions are familiar in distributed systems and security engineering, but they must be made explicit in LLM-agent architectures.

2.10 Safety, Verification, and Governance

Safety and verification are central because autonomous agents can produce effects beyond text generation. Classical MAS and autonomous-systems research has long examined model checking, runtime verification, ethical decision constraints, and verification of agent programs [33, 11, 23]. These techniques become more important when LLMs participate in action selection. A language model may produce a plausible action that violates a precondition, exceeds authority, leaks information, or conflicts with another agent’s commitment. Symbolic verification cannot eliminate all risks, but it can check well-specified properties of plans, tool calls, and execution traces.

Runtime monitoring is a practical complement to offline verification. A monitor can enforce invariants over tool calls, memory writes, inter-agent messages, and environment actions. For example, a monitor may prevent an agent from executing a destructive operation without authorization, writing to protected memory, or sending a message that violates a protocol state. Such mechanisms fit naturally with a hybrid architecture: LLMs propose and explain, symbolic controllers validate and execute, and monitors observe the boundary between proposal and action.

LLM-specific safety risks include hallucination, prompt injection, tool misuse, data

leakage, and manipulation of shared memory. Indirect prompt injection shows that external content can influence tool-using LLM applications in ways not intended by the user or developer [43]. Security analyses of MCP ecosystems emphasize that tool integration expands the attack surface for AI applications [51]. In multi-agent settings, these risks can propagate: one compromised agent may poison shared memory, issue misleading messages, or delegate unsafe actions. Therefore, governance must include identity, permissions, audit logs, provenance, and explicit boundaries between agents.

The safety literature supports a defense-in-depth position. Natural-language policies are useful for communication, but enforceable policies require structured representations. Human oversight is important, but scalable systems also need automated checks. LLM self-critique can identify some errors, but independent symbolic validation is more reliable for formal constraints. For this reason, the state of the art points toward architectures where language models are powerful but bounded participants in a larger verified control system.

Verification must also be scoped realistically. Full verification of an open-ended LLM is not currently practical. However, many safety-relevant properties concern the symbolic envelope around the model rather than the model weights themselves. A system can verify that a tool call conforms to a schema, that a plan action satisfies a precondition, that a workflow state permits a transition, that a user has authorized a capability, or that a message follows a protocol. These checks do not prove that the LLM is truthful or safe in all contexts, but they reduce the space in which model errors can produce harmful effects. The thesis can therefore pursue practical assurance by verifying the controllable parts of the agent loop.

Governance also requires traceability. A coordinated system should record why an action was selected, which agent proposed it, which module validated it, which policy authorized it, and what result followed. Such traces support debugging, evaluation, and accountability. They also support scientific analysis: without traces, it is difficult to determine whether success came from genuine coordination, accidental model knowledge, or hidden human intervention. For a master’s thesis solution, traceability is valuable both as an engineering feature and as an empirical research instrument.

2.11 Evaluation and Benchmarks

Evaluation of unified LLM-symbolic multi-agent systems requires multiple dimensions. Classical MAS evaluation considers coordination efficiency, communication overhead, task allocation quality, robustness to agent failure, scalability, and social welfare. LLM-agent evaluation considers task success, reasoning quality, tool-use accuracy, factuality, cost, latency, robustness, and safety. A unified system must be evaluated across both sets of criteria. A system that produces elegant dialogue but fails execution is inadequate; a system that executes reliably but cannot interpret natural-language goals may also fail the intended use

case.

Several benchmarks have shaped LLM-agent evaluation. AgentBench evaluates LLMs as agents across interactive tasks [68]. WebArena tests web navigation and task completion in realistic web environments [119]. ALFWorld aligns text instructions with embodied environments [97]. SWE-bench evaluates the ability to resolve real-world software issues [58]. GAIA emphasizes general assistant tasks requiring reasoning and tool use [73]. OSWorld evaluates computer-control agents in operating-system environments [114]. These benchmarks are useful, but they do not fully capture the problem of heterogeneous multi-agent coordination with symbolic verification.

Table 2.5: Evaluation dimensions for unified LLM-symbolic multi-agent systems.

Dimension	Representative questions
Task performance	Does the system achieve the specified goal, and with what cost, latency, and solution quality?
Coordination quality	Are tasks allocated effectively, messages grounded, conflicts resolved, and commitments tracked?
Symbolic validity	Do generated plans satisfy preconditions, constraints, and domain rules before execution?
Robustness	Does performance persist under rephrased goals, partial observability, agent failure, or adversarial inputs?
Safety and governance	Are tool calls authorized, logged, monitored, and constrained by explicit policies?

The evaluation gap is therefore clear. Existing benchmarks evaluate individual LLM agents, web navigation, embodied tasks, coding tasks, or general assistant behavior. Classical MAS benchmarks evaluate coordination mechanisms under formal assumptions. Fewer evaluations measure whether LLM-mediated agents can coordinate through natural language while maintaining symbolic state, formal commitments, and verified execution. This gap motivates thesis-specific scenarios that test not only whether a final task is completed, but also whether the coordination trace is interpretable, valid, and safe.

A useful evaluation design should therefore include both outcome metrics and process metrics. Outcome metrics measure whether the task was completed, how long it took, how much it cost, and whether the output met quality criteria. Process metrics measure whether the correct agent accepted the task, whether the symbolic plan was valid, whether messages followed protocol, whether tool calls were authorized, and whether failures were detected at the proper boundary. This distinction is important because an LLM system can sometimes achieve a correct result through an unsafe or non-generalizable process. Scientific evaluation should avoid rewarding such accidental success without qualification.

Evaluation should also include ablations. A unified architecture can be compared against variants that remove the symbolic planner, remove the LLM interpreter, remove protocol

tracking, or remove runtime monitoring. Such comparisons make it possible to test whether each component contributes to performance, safety, or interpretability. Without ablation, the architecture may appear effective only because the task is easy or because a single model performs most of the work. The state of the art in agent evaluation increasingly recognizes this issue, but unified LLM-symbolic MAS evaluation remains underdeveloped.

2.12 Synthesis and Research Gap

The reviewed literature shows that the components needed for unified coordination already exist, but they remain insufficiently integrated. Classical agent theory defines autonomy, interaction, commitment, planning, and verification. MAS research provides mechanisms for task allocation, negotiation, shared-state coordination, norms, and distributed constraint reasoning. Symbolic AI supplies formal representations for knowledge, plans, rules, and control. LLM research supplies linguistic interpretation, flexible communication, tool-use scaffolding, and broad task generalization. Modern interoperability protocols supply practical interfaces for tools, context, and cross-agent communication.

Table 2.6: Synthesis matrix deriving requirements for the proposed framework.

Research line	Contribution	Limitation	Derived thesis requirement
Classical agents and MAS	Autonomy, roles, task allocation, negotiation, commitments, and coordination failures.	Does not by itself handle open-ended natural-language intent or modern LLM tool scaffolds.	Treat agents as accountable participants with explicit state, capabilities, and commitments.
ACLs and interaction protocols	Performatives, protocol state, content languages, and observable communicative force.	Structured acts are often absent from free-form LLM dialogue.	Map language-mediated exchanges to typed requests, acceptances, reports, and completion evidence.
Symbolic AI	Plans, constraints, schemas, rules, and verification conditions.	Only reasons over modeled facts, actions, and contracts.	Use symbols for authorization, compatibility, evidence, and refusal rather than for all semantic interpretation.

Research line	Contribution	Limitation	Derived thesis requirement
Neuro-symbolic and LLM planning	Translation between natural language, tools, formal goals, and external solvers.	LLM-generated plans can remain plausible but invalid under action semantics.	Separate candidate proposal from feasibility checking and executable commitment.
LLM-MAS frameworks	Role-based decomposition, critique, tool use, and flexible collaboration.	Prompted roles and conversation traces do not guarantee role semantics or accountability.	Require explicit ownership, authority, traceability, and process metrics beyond final output.
Modern interoperability protocols	Tool/context access, agent discovery, task exchange, and cross-framework connectivity.	Connectivity is not coordination, and discovery is not trust.	Keep protocol transport subordinate to coordination semantics, trust decisions, and state models.
Safety and evaluation	Monitoring, authorization, audit logs, benchmark practice, and process/outcome separation.	Full verification of open-ended LLM behavior remains impractical.	Verify the symbolic envelope: contracts, permissions, transitions, traces, and bounded auxiliary capabilities.

The state of the art nevertheless leaves several unresolved issues. First, the term agent is used inconsistently across classical MAS and LLM literature, which obscures the distinction between a language model, a tool-using application, and an autonomous system. Second, LLM-based coordination often relies on free-form conversation without explicit performatives, commitments, or protocol state. Third, symbolic planning and verification are often treated as optional tools rather than as structural components of safe execution. Fourth, emerging protocols improve connectivity but do not, by themselves, solve the semantic problem of coordination. Finally, benchmark practice has not yet converged on evaluation methods for systems that combine linguistic agents, symbolic agents, tool agents, and autonomous execution components.

These gaps define the thesis positioning. A unified coordination framework should treat LLMs as linguistic reasoning modules embedded inside full autonomous agents. It should translate between natural-language interaction and symbolic coordination structures. It should distinguish proposals from commitments, and commitments from executed actions. It should expose tools through controlled interfaces, represent shared state explicitly, and validate plans before execution. Such a framework does not reject the flexibility of LLMs; rather, it

makes that flexibility usable in autonomous systems where coordination, accountability, and safety matter.

The resulting research direction is not a competition between neural and symbolic AI. It is an architectural question about how heterogeneous reasoning components should cooperate. LLMs are valuable because they make autonomous systems more communicative, adaptive, and accessible to human intent. Symbolic AI is valuable because it makes plans, constraints, commitments, and verification conditions explicit. MAS research is valuable because it explains how autonomous entities interact when no single component has all information or authority. The contribution of a unified framework is to place these elements in a coherent coordination loop, where language supports interpretation, symbols support execution semantics, and protocols support collective behavior.

Chapter 3

Research Methodology

This chapter describes the design-science method used to construct and evaluate the proposed coordination boundary. Its goal is to translate the research questions into observable units, controlled comparisons, fault scenarios, metrics, and release criteria while keeping planning quality, runtime safety, and distributed recovery as distinct forms of evidence.

The chapter is organized as follows. The study-design section defines the unit of analysis and the staged evaluation. The corpus section explains the versioned, pre-specified, and label-hidden cases. The configurations-and-fairness section defines the hybrid, direct-LLM, deterministic, and ablation controls. The local-model protocol then specifies collection, followed by the statistical measures and analysis procedure. The consensus campaign section defines the separate crash-fault evaluation, and the final sections establish provenance, reproducibility, and validity boundaries. Disclosure of AI systems used as experimental subjects or research assistants is provided in Appendix D.

3.1 Study Design

The thesis follows design science: it constructs a protocol-adaptable coordination boundary and evaluates whether the implementation exhibits the declared authorization, refusal, execution, and recovery behavior. The unit of analysis is one coordination attempt, comprising the natural-language request, admitted registry snapshot, typed request, non-authoritative proposal, symbolic predicates, authorization decision, task attempts, artifacts, and persisted trace. Outcome correctness and process correctness are analyzed separately.

The evaluation is staged. Unit and integration tests exercise individual contracts and transitions. Deterministic cases exercise the complete in-process workflow, bounded provider search, registry permutations, alternative-provider recovery, and refusal beyond the assignment limit. Container harnesses exercise A2A admission and dispatch across processes and lease-backed or consensus-backed recovery. The primary comparative study is version 0.7: 64 held-out cases arranged as 32 feasible/infeasible matched pairs, three models, two re-

quested seeds, and two independently prompted arms, producing 768 confirmatory outputs. Version 0.5 is retained as the experiment that motivated the production-path redesign; versions 0.3 and 0.4 remain earlier development evidence, and the unexecuted study-specific v0.6 protocol is superseded.

3.2 Versioned, Pre-specified, and Label-hidden Corpus

The version 0.3 corpus contains 24 base cases, four in each of six strata: direct feasible execution; decomposed feasible execution; feasible planning followed by runtime failure; missing capability or auxiliary overreach; schema, modality, graph, or ordering invalidity; and authority, typed-constraint, availability, or registry invalidity. Twelve controlled variations add four paraphrases, four availability changes, and four capability-alias perturbations. Public requests, registry snapshots, payloads, and shared proposals are stored separately from hidden reference labels.

Version 0.4 was a development cycle created after observing the version 0.3 all-refusal result. It is not retrospectively altered. Because its corpus language and bridge protocol were used to tune the redesign, it is historical rather than confirmatory evidence.

Version 0.5 is fresh and was frozen before collection. Eight domains each contribute three matched pairs, and cyclic offsets assign three pairs to each of eight invalidity categories: missing capability, unavailable agent, insufficient trust, incompatible modality, typed-constraint violation, invalid artifact contract, dependency/order error, and auxiliary overreach. The resulting 48 held-out cases are distinct from eight unscored development examples. Twelve additional runtime cases test dependency failure, timeout and unknown attempts, retry/recovery, artifact rejection, and trace behavior without entering planning accuracy. The protocol is described as *pre-specified*, not preregistered, because it was not deposited in an external timestamped registry.

Version 0.6 was drafted for the intermediate study-specific bridge but was never executed. Its manifest is preserved and marks it superseded rather than silently deleting or rewriting it.

Version 0.7 evaluates the production semantic-admission and symbolic-compilation path. A 48-case development-only corpus contains six cases in each of eight strata: paraphrase, ambiguity, negation, trust, artifact contracts, dependencies, provider recovery, and adversarial wording. It is used first with Qwen3-1.7B to expose implementation and prompt defects; all developmental runs, including failed qualifications, are retained, but their accuracy is not confirmatory evidence. The frozen corpus contains 64 held-out cases as 32 feasible/infeasible matched pairs, with four pairs in each stratum. Public cases and author-only labels are separated. The freeze records hashes of the corpus, prompt, schemas, production source, analysis, benchmark, model parameters, dependency lock, and runtime identities.

The thesis author defined the reference labels and minimum justifications before model

collection. A provenance record identifies the author, annotation date, labeling protocol, corpus version and SHA-256 digest, and declares the labels frozen. No independent adjudicator was available. Reference labels are never included in model prompts and are loaded by the scorer only after collection is complete. An automated audit checks document shape, unique identities, public/hidden alignment, hash agreement, status consistency, and justification presence; it establishes internal consistency, not independent correctness. Any post-freeze change requires a new corpus version and hash.

3.3 Configurations and Fairness

For every model, seed, and held-out case, two arms receive the same public natural-language request, admitted catalog, registry, and output-schema budget. The production hybrid arm emits a strict semantic selection that remains non-authoritative; deterministic admission constructs the typed request and the shared compiler makes the authorization decision. The direct-LLM arm emits an explicit accept/refuse decision without a plan-level symbolic authorization gate. Its decision is scored offline and is never dispatched. Both arms receive one call, temperature 0.2, top-p 1, and an 800-token limit. Invalid schema is a failed observation without semantic retry; transport and model-server failures are classified separately.

An exact-name/alias baseline admits only uniquely grounded literal catalog terms. An oracle semantic-admission baseline supplies the author-labelled semantic intent to the same deterministic compiler, and a first-provider ablation removes ranked alternative-provider recovery. These controls distinguish semantic admission from symbolic compilation and provider selection. Registry-order permutations and the 4,096-assignment refusal boundary are evaluated deterministically.

Planning-only decisions and runtime compliance are not conflated. Dependency-gating and durable-trace ablations may leave the planning decision unchanged while altering execution safety or observability. Those mechanisms are therefore scored through dispatch, skip, duplicate-effect, and trace-completeness measures rather than through feasibility accuracy alone.

The runtime-only study contains five dependency scenarios, five auxiliary-admission scenarios, and five trace-evidence scenarios, each repeated ten times under the applicable strict and negative-control configurations. This produces 350 deterministic observations in a separate immutable bundle. The dependency control bypasses predecessor status; the auxiliary controls disable the feature or bypass its bounds; and the trace control suppresses returned diagnostic evidence while preserving state-critical persistence. These rows are not treated as LLM planning observations.

3.4 Local Model Protocol

The fixed local models are qwen/qwen3-1.7b, google/gemma-4-e2b, and qwen/qwen3-8b. LM Studio is addressed at `http://127.0.0.1:1234/v1` with a non-secret local API-key value. Temperature is 0.2, top-p is 1, the maximum response is 800 tokens, and requested seeds are 11 and 29. Seed 11 is primary and seed 29 is a stability replication rather than an independent sample.

All other models are unloaded before a batch. The runner loads one exact identifier, uses 16,384-token context and parallelism one, and verifies the model catalog. Qwen3-1.7B is collected first. Only runtime completeness and artifact integrity are checked before Gemma-4-E2B and Qwen3-8B run under the identical protocol; poor Qwen performance cannot trigger tuning or stop the sequence. For each of 64 cases, both arms are collected at two requested seeds for each of three models, yielding exactly 768 immutable confirmatory outputs. Every record preserves arm/model/seed/case identity, prompt, raw response, parsed object or schema failure, latency, token counts, and call count. Interrupted runs may resume only after corpus, protocol-hash, and existing-record identity checks; completed case files are never rewritten and scoring is blocked until the complete matrix exists.

For version 0.4 the complete admitted `ProblemRequest` and registry snapshot are public prompt inputs, while the output is only a `LinguisticPlanDraft`. Deterministic hydration occurs after model output. When the draft contains repairable structural errors—such as omitted, duplicate, or unknown requirement references or invalid dependencies—the model receives those public errors once and may return one repaired draft. Capability unavailability and symbolic infeasibility do not trigger repair. Initial linguistic latency, repair latency, symbolic-validation latency, execution latency, and total latency are stored separately.

3.5 Qwen-First Development and Protocol Freeze

Development begins with Qwen3-1.7B because it is the least expensive model in the fixed set and provides broad feedback before larger-model collection. Its 48 development cases exercise catalog-size variation, registry permutations, missing providers, malformed intent, alias collisions, prompt injection, ambiguous and negated language, and search exhaustion under both the hybrid and direct arms at seeds 11 and 29. Prompt and implementation corrections are permitted in this phase. Qualification requires complete output accounting, no unhandled exceptions, fail-closed ambiguous or invalid input, and 100% deterministic compiler invariants. Developmental accuracy is reported only as process history.

After qualification, the protocol lock freezes production source hashes, corpus and labels, prompts, schemas, model parameters, analysis and benchmark code, success criteria, dependency lock, and runtime identity. Any behavioral correction after this point requires

a new version and a complete three-model rerun. The author-only labels are disclosed as a limitation rather than described as independent ground truth.

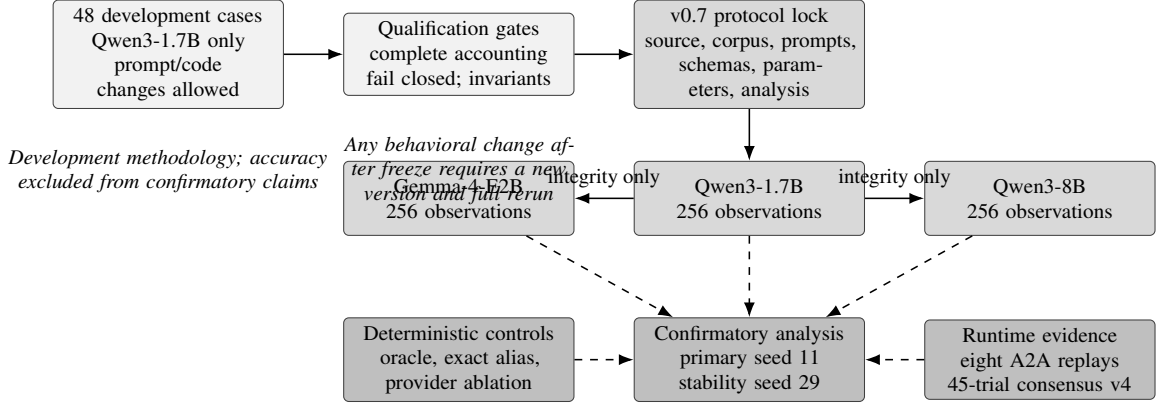


Figure 3.1: Qwen-first development, protocol freeze, and separated confirmatory evidence tracks.

3.6 Metrics and Statistical Analysis

Primary semantic measures are field-level accuracy and F1. Decision measures are balanced accuracy, unsafe acceptance, feasible recall, false refusal, plan validity, provider recovery, model/category errors, and seed agreement. Efficiency measures include latency, calls, prompt tokens, and completion tokens. Deterministic and distributed measures include topological-order and registry-permutation invariants, bounded-search refusal, authorization before dispatch, duplicate dispatch, repeated effects, stale-operation rejection, crash recovery, and terminal correctness.

The 32 matched pairs are the primary statistical units; model and requested-seed outputs are repeated measurements. Seed 11 supplies the primary estimates, while seed 29 measures stability. Matched-pair clustered bootstrap intervals are reported for hybrid-minus-direct unsafe acceptance and feasible recall, and for hybrid-minus-exact-alias balanced accuracy. Results are reported separately for every model and category rather than pooled into a single model-independent claim. Functional results and artifact identities are hashable; wall-clock latency remains hardware-dependent evidence.

The predeclared gates are zero primary-seed unsafe hybrid acceptances; at least 75% feasible recall overall and 60% for every model; safety superiority over the direct LLM; feasible-recall non-inferiority within ten percentage points; superiority over the exact-name/alias baseline; and 100% oracle/compiler invariants. Every gate is reported, including negative outcomes, and no failed gate is tuned away after freeze.

3.7 Version 0.8 Follow-up Protocol

Version 0.8 uses 72 development cases and 96 held-out confirmatory cases. Each phase contains feasible/infeasible matched pairs in twelve categories: low-overlap paraphrase, disjunction, negation, quoted adversarial text, unknown required entities, trust, artifact contracts, dependencies, multi-goal options, provider recovery, Unicode/accents, and Spanish paraphrase. The automated overlap audit requires at least 75% of confirmatory goal expressions to lack an exact normalized identifier, name, or alias substring. Public cases and hidden labels are physically separated.

Only qwen/qwen3-1.7b and google/gemma-4-e2b are allowed in v0.8. Qwen3-8B remains part of the unchanged v0.7 evidence but is rejected by every v0.8 runner and excluded from v0.8 schemas, locks, analyses, and claims. Both arms run at requested seeds 11 and 29. A model batch begins by checking lms, unloading other models, loading the exact identifier, and verifying the OpenAI-compatible advertised identity. Raw response, schema, prompt, timing, token use, runtime failure, and returned model identity are preserved for every observation.

Development requires exactly $72 \times 2 \times 2 \times 2 = 576$ observations. Each model must have zero unsafe hybrid acceptances, at least 65% hybrid feasible recall, derived-status macro-F1 of at least 70%, at least 99% schema-valid output, no category with zero feasible recall, and all symbolic invariants passing. Prompt or implementation revision is allowed only against development evidence, and all unsuccessful iterations are retained.

The confirmatory phase remains closed until two output-blind author labeling passes are completed at different times and disagreements are reconciled. This process reduces accidental inconsistency but remains author labeling, not independent adjudication. After that review, the lock freezes source, corpus, prompt, schema, model, seed, arm, gate, and identity hashes. The required confirmatory matrix is $96 \times 2 \times 2 \times 2 = 768$ immutable observations, with no semantic repair or selective retries. Raw interpretation, canonical intent, admitted request, and authorized plan are scored separately, and direct proposals pass through the shared FeasibilityAnalyzer. Cross-version comparisons are descriptive because both corpus and semantic interface changed.

3.8 Consensus Crash-Fault Campaign

RQ5 is evaluated as a separate core track rather than as additional LLM observations. Before accepted execution, the campaign protocol fixes three fresh trials for clean formation at voter targets three, five, and seven; live transitions $3 \rightarrow 5 \rightarrow 3$ and $5 \rightarrow 7 \rightarrow 5$; leader termination; minority partition; majority loss and restoration; persistent restart; failed-voter replacement; unavailable audit projection; concurrent ownership; and crashes after attempt start, after

external dispatch, and during aggregation. Every trial uses a new Docker Compose project, new volumes except where persistence is the object of the test, and a non-existing evidence directory.

Safety observations are one committed session owner, monotonic fencing, rejection of stale writes, attempt start before completion, at most one terminal result, refusal of authoritative mutation without quorum, serialized membership changes, and zero repeated external effects when the receiving fixture enforces operation keys and fences. Conditional liveness requires leader-failure progress within 30 seconds while a majority survives, quorum-loss detection within 10 seconds, restored progress within 30 seconds, and membership convergence within 120 seconds. Steady state additionally requires authoritative role agreement and no pending membership operation. Repetition tests implementation stability but does not create independent deployment populations.

The campaign invokes etcd 3.6.13 as the Raft implementation; it does not reimplement or re-prove Raft. The fault model is crash/omission under eventual synchrony with a surviving majority. All containers share one Docker host, so the campaign does not represent independent machines, disks, regions, switches, or Byzantine behavior.

3.9 Provenance and Reproducibility

Each run directory records Git SHA, dirty-state flag, corpus hash, model ID, prompt/schema version, seed, dependency-lock hash, Docker image digest, Python, PostgreSQL, A2A and LM Studio versions, operating system, CPU, RAM, and available accelerator information. Reports are written to new run-ID directories; convenience summaries are derived and may be regenerated, but the only copy of a raw run is never overwritten.

The release gates require all unit and integration tests, static lint and type checks, the deterministic compiler benchmark, eight fixture-A2A replays, the PostgreSQL and upstream-derived A2A checks, the full consensus campaign, protocol and evidence-schema checks, and LuaLaTeX/BibTeX compilation. Coverage thresholds are 85% overall production code; 90% for semantic admission, symbolic compilation, feasibility, coordination execution, trace validation, plan hydration, and evidence preflight; 85% for cluster discovery, launcher, etcd, registry, service, and SDK modules; and 80% for PostgreSQL, the consensus campaign driver, and current evaluation modules. Fixture servers and system drivers are exercised as subprocesses or containers rather than counted in the production aggregate. A release cannot be frozen while any threshold, clean-source evidence, institutional metadata, evaluated-source commit, or packaging-commit field is open.

3.10 Validity Boundaries

The corpus is designed and labelled by the thesis author rather than sampled from a deployment population or independently adjudicated. This creates researcher-bias, construct-circularity, and baseline-design threats: agreement measures conformance to the declared framework criteria, not neutral benchmark truth. Freezing labels before collection and hiding them from prompts reduces leakage but does not create independence. The agents are mostly controlled fixtures, and the local models are small. Contract validity does not establish semantic truth. The PostgreSQL lease-and-fencing evidence concerns measured crash faults and is not consensus evidence. The etcd campaign is a core implementation evaluation but remains a single-host, project-authored fault campaign rather than proof of all Raft properties or multi-host availability. A2A admission establishes local origin, transport, trust, and credential policy; it does not establish cryptographic identity or malicious-agent resistance. These boundaries constrain every answer to the research questions.

Chapter 4

Proposed Theoretical Framework

This chapter develops the theoretical framework proposed by the thesis. The framework addresses a specific coordination problem: how a distributed collection of heterogeneous agents can accept a problem stated in natural language, establish whether the problem lies within its current operational boundary, and construct a coordinated solution without allowing linguistic plausibility to substitute for executable justification. The proposed answer combines a central coordinating authority with distributed execution, capability-based admission, explicit commitments, and a symbolic feasibility decision. Language-oriented components interpret and propose; symbolic components authorize, record, and reject when the available evidence is insufficient.

The framework is intentionally independent of a particular communication protocol, model provider, programming language, or deployment platform. Chapter 5 instantiates the concepts using concrete technologies, but those choices are not part of the theoretical claim. The present chapter instead defines the objects, boundaries, and properties that any conforming implementation must preserve.

The goal of the chapter is to state precisely what information and predicates separate a plausible proposal from an authorizable plan. It first identifies the purpose, assumptions, and design requirements of the framework. It then defines agents, capabilities, linguistic mediation, bounded auxiliary capabilities, commitments, communication properties, and feasibility. The subsequent sections state the framework properties, exercise them in a worked example, delimit the theory, and finally specify the safety and conditional-liveness obligations introduced when the one logical authorization role is replicated through consensus.

4.1 Purpose and Contribution

Open multi-agent systems derive value from heterogeneity. Agents can differ in ownership, internal architecture, reasoning method, data access, and operational authority while still contributing to a shared objective. The same heterogeneity also complicates coordination.

A fluent description of a capability does not establish that the capability exists; a plausible decomposition does not establish that its subtasks can exchange compatible artifacts; and a successful delegation message does not establish that the delegated work has been completed. These distinctions are especially important when large language models participate in interpretation and planning, because linguistic confidence can conceal missing capabilities or violated constraints.

The framework treats this problem as one of *controlled openness*. Participation is open in the sense that independently implemented agents can be admitted through externally visible contracts. It is controlled because admission, authority, capability coverage, and plan acceptance remain explicit decisions. The operational boundary is therefore neither the set of tasks an LLM can describe nor the set of outcomes that appear plausible in conversation. It is the set of task assignments that can be justified from the currently admitted agents, approved auxiliary capabilities, compatible data flows, and applicable authority constraints.

The principal contribution is a disciplined boundary between linguistic semantic selection and symbolic authorization. Natural-language reasoning is used where ambiguity and semantic comparison make it valuable: selecting admitted goal, trust-policy, artifact-contract, and exclusion identifiers from a finite catalog, or reporting unresolved terms. Once those semantics are admitted, deterministic components own dependency closure, typed-request construction, provider assignment, schemas, modalities, trust constraints, completion evidence, lifecycle rules, and authorization. The result of coordination is total at the planning boundary: either an accepted plan is produced with an auditable justification, or a structured infeasibility report identifies why no plan can presently be authorized.

Figure 4.1 summarizes this separation. The central coordinating authority is a logical decision role rather than a requirement that all execution occur in one process. Participating agents remain distributed and retain control of their internal operation. Centralization applies to the acceptance of the shared plan and the maintenance of its coordination state. A fault-tolerant implementation may therefore replicate coordinator processes or move state into durable storage, provided that each coordination session still has one authoritative plan state and one accepted authorization decision.

4.2 Assumptions and Design Requirements

The framework rests on five assumptions. First, participating agents are treated as black boxes with respect to their internal reasoning. An agent may use rules, a planner, a language model, retrieval, statistical learning, deterministic software, or a hybrid of these methods. Coordination depends only on the contract the agent exposes and on the evidence it returns. Second, the registry is evaluated as a session-level snapshot. Agents may join, leave, or change between sessions, but a feasibility decision is made against a definite set of available

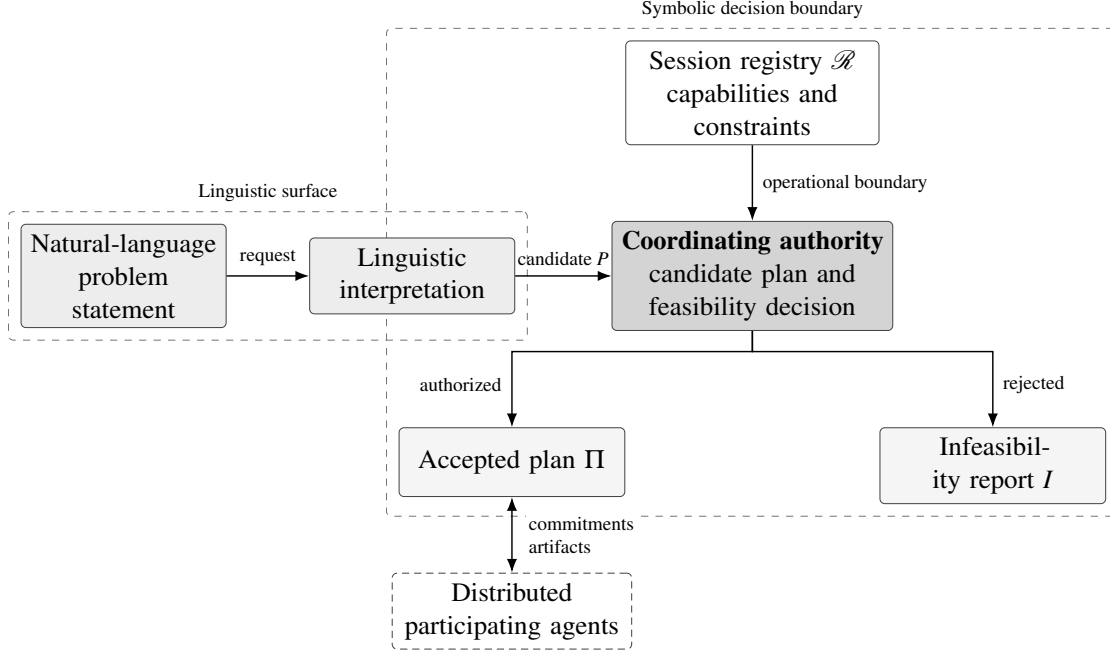


Figure 4.1: Conceptual boundary of the proposed framework. Linguistic interpretation produces a candidate problem representation; the coordinating authority evaluates it against the session registry and returns either an authorized plan or an explicit infeasibility report. Execution remains distributed among participating agents.

descriptions and states.

Third, capability descriptions and admission metadata are assumed to be sufficiently honest for contractual reasoning. This is not an assumption that agents are infallible. It means that the coordinator may use an admitted capability claim as evidence when constructing a plan. Misrepresentation, compromised identities, and adversarial capability advertisements require trust-establishment mechanisms beyond the present theory. Fourth, the central coordinator is trusted to apply the declared checks correctly. The framework constrains this role through explicit inputs, outputs, and traces, but it does not solve Byzantine coordination among competing coordinators. Fifth, deterministic validation is limited to properties that can be expressed in a contract. Schema validity, provenance, bounds, permissions, and required fields may be decidable even when the semantic quality of an open-ended result is not.

Within those assumptions, the design is governed by nine requirements. *Openness* permits independently developed agents to enter or leave through explicit admission procedures. *Vendor independence* prevents coordination semantics from depending on an agent’s internal framework or model provider. *Interoperability* requires shared representations for capabilities, tasks, status, artifacts, and errors. *Distributed operation* keeps participating agents independently deployable and avoids shared hidden state as a prerequisite for cooperation. *Coordinated authority* assigns responsibility for the global feasibility decision and the accepted plan. *Simplicity* prefers direct delegation over decomposition and deterministic checks

over discretionary linguistic judgments when both can decide the same condition. *Observability* records decisions, commitments, execution events, and artifacts. *Fault containment* treats timeouts, unavailable agents, invalid artifacts, and rejected plans as explicit states rather than exceptional conversational endings. Finally, *bounded authority* prevents either the coordinator or a generated auxiliary capability from acquiring permissions merely because a request would benefit from them.

These requirements create a deliberate asymmetry. Execution is distributed, but authorization of the shared plan is centralized. The asymmetry is not asserted as a universal necessity for all multi-agent systems; decentralized negotiation and consensus are valid alternatives. It is a design choice for this framework because a single decision role provides a definite answer to the question the thesis studies: whether a requested solution is justified within the current system boundary.

The scope is correspondingly limited. The framework does not prove the correctness of an agent's internal reasoning, guarantee that an accepted plan will succeed at runtime, optimize resource allocation in competitive markets, or define decentralized negotiation among competing authorization authorities. A concrete implementation may use consensus to replicate the state of this one logical role without changing the theory. The framework also does not replace an agent communication language or transport protocol. Its claim is narrower: under the stated assumptions, it makes the basis for plan acceptance, delegation, and refusal explicit and testable.

4.3 Conceptual Model of Agents and Capabilities

The conceptual model separates an agent's implementation from its externally relevant coordination state. This permits a language-model agent, a symbolic planner, a database service, and a deterministic program to participate under the same abstraction without implying that their internal forms of autonomy are identical.

Definition 4.1 (Agent). An agent is a tuple

$$a = (id, C(a), M_{in}(a), M_{out}(a), \tau(a), \sigma(a)),$$

where id is a unique identity, $C(a)$ is a finite set of declared capabilities, $M_{in}(a)$ and $M_{out}(a)$ are accepted input and produced output modalities, $\tau(a)$ is an admitted trust or authority level drawn from a partial order, and $\sigma(a)$ is the agent's observable availability state for the current session.

Identity and availability are included because capability alone is insufficient for delegation. An agent that advertises the correct transformation but is unavailable, unauthenticated, or

unauthorized for the requested context cannot justify a feasible assignment. The trust value $\tau(a)$ is intentionally abstract: an implementation may derive it from credentials, local policy, organizational membership, or another admission mechanism.

Definition 4.2 (Capability contract). A capability contract is a tuple

$$c = (\textit{name}, \Sigma_{\text{in}}, \Sigma_{\text{out}}, \gamma, e, \nu),$$

where Σ_{in} and Σ_{out} are input and output schemas, γ is a set of preconditions and authority constraints, e specifies the evidence expected on completion, and ν is a deterministic contract validator for that evidence and the produced artifact.

The validator ν establishes conformance to the declared contract, not universal semantic correctness. For a numerical service, it may recompute a result; for a document generator, it may only verify format, provenance, required sections, and declared bounds. This distinction prevents the theory from claiming that every open-ended capability has a complete oracle. Where semantic correctness is undecidable or expensive, the contract must state the evidence that can actually be checked and leave residual uncertainty visible.

Capability matching is represented by the relation $a \models q$, meaning that at least one capability in $C(a)$ covers requirement descriptor q under the applicable schemas, modalities, and constraints. Coverage may be established by exact symbolic matching or proposed by semantic comparison and then confirmed against explicit contract fields. A linguistic similarity score alone is not sufficient to establish $a \models q$.

Definition 4.3 (Registry and operational boundary). Let $\mathcal{R}$ be the registry snapshot for a coordination session. Its capability space is

$$K(\mathcal{R}) = \bigcup_{a \in \mathcal{R}} C(a),$$

and its supported modalities are

$$M(\mathcal{R}) = \bigcup_{a \in \mathcal{R}} (M_{\text{in}}(a) \cup M_{\text{out}}(a)).$$

The operational boundary $\mathcal{B}(\mathcal{R})$ is the set of agent–capability pairs (a, c) for which $a \in \mathcal{R}$, $\sigma(a)$ is available, $c \in C(a)$, and the admission and authority constraints applicable to c are satisfied.

The boundary is evaluated rather than presumed. It may contract when an agent becomes unavailable or loses authority, and it may expand when a new agent is admitted. Task-local auxiliary capabilities can extend it only under the bounded rules defined later in this chapter.

The boundary therefore describes what the system can justify at a particular decision point, not what it might be able to learn or deploy in the future.

4.4 Linguistic Agents and Symbolic Mediation

A linguistic agent is an agent whose principal capabilities transform, interpret, or produce language-mediated artifacts. Typical capabilities include intent interpretation, summarization, classification, translation, information extraction, semantic matching, question answering, and natural-language explanation. Such agents are necessary because user goals and capability descriptions frequently arrive as prose, whereas coordination requires structured task and artifact contracts.

Definition 4.4 (Linguistic agent). An agent a is linguistic with respect to a capability $c \in C(a)$ when the contract for c accepts or produces natural language and the capability’s primary function is to mediate between linguistic meaning and another linguistic or structured representation.

This definition concerns function rather than implementation. A linguistic agent may use a large language model, a smaller statistical model, rules, retrieval, a controlled vocabulary, or a hybrid pipeline. Broad contextual interpretation may justify an LLM, whereas language detection, constrained extraction, or normalization may be implemented more simply. Treating every language operation as an LLM task would weaken the simplicity requirement and could increase cost, latency, and uncertainty without adding relevant capability.

The linguistic–symbolic bridge is consequently modeled as a sequence of artifacts and validation boundaries, not as a single translation component. The theory requires that language-mediated semantic selection, deterministic admission and compilation, symbolic authorization, and feedback grounding remain distinguishable even when a concrete implementation realizes them through the same software runtime. Four transitions define the core loop.

Transition T1: semantic selection. A user expression u is transformed into a strict, non-executable semantic intent $\hat{S}$ over identifiers admitted by an authoritative catalog. Every wire field is required; unresolved or ambiguous language remains explicit. A single repair may correct only JSON syntax or schema conformance.

Transition T2: admission and compilation. Deterministic grounding validates $\hat{S}$, applies explicit negations and exclusions, computes dependency closure, and constructs the typed request P from authoritative trust, artifact, schema, modality, side-effect, and validation contracts. A bounded symbolic compiler then considers all compatible providers

and constructs task identity, dependencies, execution order, artifacts, validation, and completion contracts.

Transition T3: authorization. Symbolic predicates evaluate the compiled plan against capability coverage, data-flow compatibility, dependencies, availability, authority, fencing, exclusions, and evidence contracts. The result is an accepted plan Π or an infeasibility report I .

Transition T4: feedback grounding. Status updates, artifacts, timeouts, invalid outputs, and protocol errors become structured observations linked to the corresponding task and commitment. These observations may discharge a commitment, trigger bounded replanning, or terminate the plan with an execution failure.

Between T1 and T2, names are grounded only through the catalog’s stable identifiers and declared aliases. Deterministic canonicalization preserves explicit prohibitions, applies declared defaults, removes redundant prerequisite goals already implied by a terminal goal, and refuses when a forbidden capability is required by dependency closure. Ambiguous or unresolved names remain interpretation evidence and cause refusal; they do not become hidden capabilities. Typed callers that already provide a valid P bypass linguistic interpretation entirely.

Natural-language explanations may be rendered from either Π or I , but the explanation is not the authoritative state. The symbolic artifact remains available for inspection and execution. Figure 4.2 shows the flow across the three conceptual lanes.

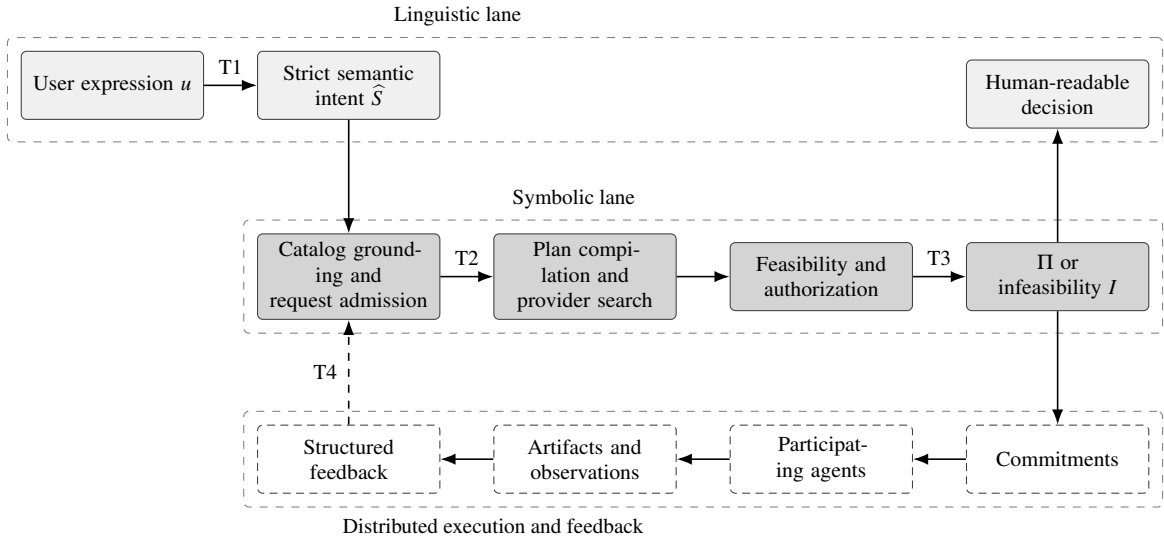


Figure 4.2: Artifact flow across the linguistic, symbolic, and distributed-execution lanes. Linguistic components select strict semantic identifiers; deterministic admission, compilation, provider search, and validation authorize a plan or record infeasibility; execution observations return through a structured feedback path.

The framework is compatible with, but not equivalent to, classical BDI terminology. The admitted problem contributes to the coordinator’s beliefs, a candidate plan represents a proposed course of action, and authorization establishes an intention-like commitment to execution [87, 88]. This is an architectural analogy rather than a claim that the tuples defined here constitute a complete BDI semantics. Similarly, structured delegations and status reports can carry communicative forces related to request, inform, and commit [6, 93, 17, 34], but their exact message encoding belongs to the implementation layer.

4.5 Bounded Auxiliary Linguistic Capabilities

An admitted registry will not contain every narrow language transformation that a decomposition may require. Rejecting an otherwise feasible plan because a trivial normalization or extraction step is absent can be disproportionate; assigning the same step to a general-purpose LLM can also be unnecessarily complex. The framework therefore permits a task-local auxiliary linguistic capability when the missing operation can be constructed from an approved, bounded method and validated against an explicit contract.

Auxiliary synthesis does not mean unrestricted generation of executable agents, arbitrary code, or newly trained models. It means constructing a temporary capability specification by selecting or composing methods from an admissible family and binding their parameters to the current task. The resulting logical agent may be realized by an existing sandbox, rule engine, model endpoint, or controlled execution service in the implementation. The theory requires the specification and its bounds; it does not prescribe the runtime that realizes them.

Definition 4.5 (Auxiliary capability specification). An auxiliary capability specification is a tuple

$$x = (\text{purpose}, \Sigma_{\text{in}}, \Sigma_{\text{out}}, m, v, b, \lambda),$$

where m is a selected method from an approved family, v is its deterministic contract validator, b states resource and authority bounds, and λ binds the specification to the lifecycle of one coordination plan.

Four rules govern admission of an auxiliary specification.

Rule A1 (Approved construction). The method m must be selected from, or composed entirely from, an approved finite family whose operational semantics are known. Synthesis may bind parameters and connect approved steps; it may not introduce arbitrary executable code or an unreviewed model.

Rule A2 (Contract validation). The input schema, output schema, expected evidence, and deterministic validator v must be available before the auxiliary capability is authorized. Failure of v is an execution failure, not a linguistically negotiable result.

Rule A3 (Bounded effects and authority). The capability must be side-effect free or restricted to effects explicitly authorized by b . It may not acquire persistent write access, external credentials, or privileges that are absent from the accepted plan.

Rule A4 (Lifecycle and provenance). The lifecycle token λ limits the auxiliary capability to its plan, and every invocation records the selected method, parameters, inputs, outputs, and validation result. The specification does not enter the persistent registry automatically.

These rules distinguish a bounded micro-capability from an attempt to synthesize expertise. Extracting fields from a fixed schema, normalizing a constrained identifier, or applying an approved classifier may be eligible. Producing a medical diagnosis, forecasting an unconstrained market, or generating privileged database operations is not eligible merely because a language model can describe a procedure. When no registered capability and no admissible auxiliary specification covers a requirement, the correct planning result is infeasibility.

4.6 Coordination, Commitments, and Communication Abstraction

The framework uses hierarchical coordination in a limited sense. The coordinator owns interpretation state, the registry snapshot, candidate-plan evaluation, and acceptance of the joint plan. Participating agents retain autonomy over whether and how they execute a delegated capability, subject to their advertised contracts. The hierarchy therefore establishes responsibility for the global decision without converting distributed agents into internal modules of the coordinator.

Central coordination is selected here because local views are insufficient to decide joint feasibility. An individual agent may know that it can perform one transformation without knowing whether another agent can consume its output or whether the complete artifact set satisfies the user's goal. The coordinator maintains the cross-agent dependency graph and the evidence required to justify the overall proposal. This choice introduces a concentration of trust and a potential availability bottleneck, which are acknowledged as limitations rather than presented as universal properties of open systems. Crash fault tolerance can be added without changing the theoretical role by replicating session state through a consensus-backed store and fencing ownership so another coordinator instance can resume after failure. Consensus here protects the state of one logical authority; it does not create negotiation among independent authorities. Byzantine coordination among mutually suspicious or competing coordinators remains outside this framework.

Definition 4.6 (Commitment). A commitment is a tuple

$$\mu = (d, r, t, e, \kappa_t, s),$$

where d is the debtor agent, r is the coordinating creditor acting on behalf of the requester, t is the delegated task, e is the expected completion evidence, κ_t is the discharge condition, and s is the observable commitment state.

A commitment is created only after plan authorization and delegation acceptance. It is discharged when the expected artifact and evidence satisfy the task contract, or terminated with a recorded failure. Maintaining commitments as symbolic state addresses a weakness of conversation-only coordination: dialogue can indicate willingness, but it does not by itself establish who owes which artifact or what observation counts as completion [98].

4.6.1 Required Communication Properties

The communication layer transports coordination artifacts but does not decide their meaning. To preserve the theory, an implementation must provide authenticated addressability, correlated task delegation, capability and contract conveyance, status and artifact delivery, timeout and cancellation semantics, and detectable failures. Messages associated with a session must carry enough identity and correlation information to reconstruct their causal order even when the underlying transport does not guarantee global ordering.

Capability descriptions must be versioned or otherwise bound to the registry snapshot used for feasibility. Delegations must identify the task, plan, expected artifact, and applicable lifecycle. Results must identify their producing agent and task so that validation and commitment discharge are unambiguous. Duplicate, delayed, or out-of-order messages must be detectable and handled idempotently at the coordination layer. Lack of progress must become an observable timeout rather than an assumption that the agent remains active.

These requirements define a communication abstraction, not a specific protocol. Any protocol or composition of protocols may instantiate it if the required artifacts and observations can be represented without exposing agent internals. The abstraction may be implemented with different transports, local runtimes, or linguistic runtimes, provided that each exposes the same capability, task, artifact, and trace contracts to the feasibility model. Planning, feasibility, and authorization remain coordination-layer responsibilities rather than transport-layer responsibilities. Substitutability is therefore conditional: changing the transport or runtime must not change the semantics of capability admission, task correlation, commitment state, or feasibility evidence.

4.7 Feasibility Model

4.7.1 Evidence-grounded semantic admission in version 0.8

Version 0.8 refines the linguistic–symbolic boundary without granting the model planning authority. Candidate retrieval collects every exact identifier, name, and alias match, adds five lexical BM25/TF–IDF candidates, and expands dependency ancestors. Retrieval only reduces the prompt search space: descriptions influence ranking but cannot authorize a goal. Every selected goal, policy, contract, or exclusion must point to exact request evidence outside ignored quoted spans. Zero terminal goals are valid when no safe grounding exists.

Ambiguity is no longer a model-authored verdict. The linguistic component reports alternatives and required unknown terms; deterministic admission derives *resolved*, *ambiguous*, or *unresolved* after checking executable disjunctions, negation scope, quotation boundaries, exclusions, and evidence consistency. Canonicalization may normalize identifiers, apply defaults, remove redundant prerequisites, and reject contradictions, but it may not add a goal. Thus lexical similarity proposes a bounded choice while authorization still requires symbolic identity and contract checks.

Feasibility is a pre-execution judgment about whether a plan is justified by the current contracts and constraints. It is not a prediction that every agent will execute successfully. Runtime failure remains possible and is handled through commitment state and structured feedback. This separation is essential: without it, a failed execution could be mistaken for an unsound planning decision, or an optimistic planning decision could be defended by a later successful accident.

Definition 4.7 (Problem request). A problem request is a tuple

$$P = (g, \mathcal{Q}, \chi, \mathcal{O}, \Gamma),$$

where g is the goal, $\mathcal{Q}$ is a finite set of required capability descriptors, χ is contextual information, $\mathcal{O}$ is the required artifact set, and Γ is the set of explicit constraints. A descriptor in $\mathcal{Q}$ may be marked as auxiliary-eligible, but eligibility is confirmed only by Rules A1–A4.

Definition 4.8 (Coordination plan). A coordination plan is a tuple

$$\Pi = (\mathcal{T}, q, \alpha, \prec, \mathcal{X}_\Pi, \mathcal{A}_{\text{out}}, \kappa),$$

where $\mathcal{T}$ is a finite set of tasks, $q : \mathcal{T} \rightarrow \mathcal{Q}$ maps each task to its capability requirement, α assigns each task to a registered agent or an approved auxiliary specification, $\prec$ is a dependency relation, $\mathcal{X}_\Pi$ is the finite set of auxiliary specifications, $\mathcal{A}_{\text{out}}$ is the expected artifact set, and κ is the plan completion criterion.

For an auxiliary specification x , the notation $x \models q(t)$ means that its approved method and contract cover the requirement of task t . For a registered agent, coverage retains the meaning established earlier. The executor assignment is total when every task has exactly one accountable executor at authorization time; redundancy or collaborative subtasks can be represented explicitly as additional tasks rather than as an ambiguous assignment.

Definition 4.9 (Feasibility). Given a request P and registry snapshot $\mathcal{R}$, feasibility holds when there exists a plan Π such that

$$\text{Feasible}(P, \mathcal{R}) \iff \exists \Pi [\text{Complete}(\Pi, P) \wedge \text{Covered}(\Pi, \mathcal{R}) \wedge \text{Compatible}(\Pi) \\ \wedge \text{Acyclic}(\prec) \wedge \text{Authorized}(\Pi, \mathcal{R}) \wedge \text{Verifiable}(\Pi)].$$

These predicates are operational acceptance criteria for implementation and evaluation, not a proof of the internal correctness of arbitrary agents. They specify which evidence must be present in the coordinator's trace before a plan may be authorized. The claim is therefore limited to pre-execution justification under the admitted contracts and registry snapshot.

The predicates make the acceptance criteria explicit. Complete requires that the task set and expected artifacts cover $\mathcal{Q}$ and $\mathcal{O}$. Covered requires

$$\forall t \in \mathcal{T}, \quad \alpha(t) \models q(t),$$

with every registered assignment drawn from $\mathcal{B}(\mathcal{R})$ and justified by a recorded capability contract, and every auxiliary assignment drawn from $\mathcal{X}_\Pi$ after satisfying Rules A1–A4. Compatible requires each dependency edge to connect recorded output and input schemas, modalities, and artifact identifiers, or an explicitly admitted bridge with its own validator. Acyclic requires $\prec$ to define a valid finite partial order. Authorized checks availability, trust, permissions, resource limits, and the constraints in Γ against the registry snapshot and policy evidence used for the decision. Verifiable requires a completion-evidence contract and deterministic validator for every task and requires κ to be decidable from recorded task states, artifacts, and validation results.

An infeasibility report is a structured artifact $I = (\text{code}, U, E, H)$, where U identifies unsatisfied requirements or constraints, E records the checks that produced the verdict, and H contains admissible remediation hints, such as admitting a missing capability or providing a required modality. A remediation hint is not a fabricated plan. It explains how the boundary would need to change before the request could be reconsidered.

The operational boundary is dynamic across sessions but fixed for a single authorization decision. If the registry changes during planning, the candidate must be re-evaluated against the new snapshot. If it changes after authorization, the affected commitment becomes unavailable or invalid and enters the feedback loop. This rule avoids accepting a plan from a capability view that no longer corresponds to the agents available for execution.

4.8 Properties of the Framework

The following properties state the behavior expected from a conforming implementation. They are conditional on the assumptions and contract descriptions introduced above; they are not claims about the internal correctness of arbitrary agents.

Property 4.1 (Justified plan acceptance). If the coordinator accepts Π , every task has an accountable executor whose admitted contract covers its requirement, every dependency is compatible, and every explicit authority and validation condition used by Definition 4.9 has passed. Acceptance does not imply that execution cannot subsequently fail.

Property 4.2 (Simplicity preference). If one admitted agent can satisfy the complete request under the same constraints and evidence requirements, the coordinator selects direct delegation before considering decomposition. Among evaluated multi-agent candidates, it prefers fewer tasks, fewer auxiliary specifications, and a shallower dependency graph, in that order, unless an explicit request constraint establishes another priority.

Property 4.3 (Bounded auxiliary synthesis). For every accepted plan, $\mathcal{X}_\Pi$ is finite; each member satisfies Rules A1–A4; and no member remains valid after its lifecycle token expires.

Property 4.4 (Traceability). Every decision can be reconstructed from the request representation, registry snapshot, candidate plan, predicate results, selected assignments, commitments, validation evidence, and execution observations. The trace distinguishes planning infeasibility, coordination failure, agent failure, and artifact-validation failure.

Property 4.5 (Explicit refusal). At the planning boundary, the coordinator returns either an accepted plan or an infeasibility report. Missing coverage, authority, compatibility, or evidence cannot be replaced by an unsupported linguistic assertion of feasibility.

These properties provide evaluation targets rather than completed proofs. The implementation must operationalize each predicate, and the empirical evaluation must examine whether observed traces conform to the stated behavior.

4.9 Worked Example

Consider a registry snapshot containing three agents. Agent a_1 summarizes English documents and returns a textual summary with source references. Agent a_2 computes arithmetic expressions over typed numeric inputs and returns both a value and a reproducible calculation trace. Agent a_3 executes read-only aggregate queries expressed through a constrained query object and returns a tabular artifact. All three agents are available and admitted for the relevant data.

The user requests: “Summarize the quarterly report, calculate the percentage change in revenue from Q1 to Q2, and list the five customers with the highest total order value.” Interpretation produces a candidate request whose required capabilities are document summarization, extraction of two named quarterly values, percentage calculation, and constrained aggregation. The required artifacts are a summary, a percentage, and a customer table.

Direct delegation fails because no single agent covers the complete request. The summarization, arithmetic, and aggregation requirements are covered by a_1 , a_2 , and a_3 , respectively. The registry lacks the narrow transformation that extracts the Q1 and Q2 revenue cells from a table with a known schema. This gap is auxiliary-eligible: an approved column-selection method can be parameterized with the two quarter labels, its output schema is a pair of currency values, and its validator checks column identity, numeric type, provenance, and cardinality. The coordinator therefore proposes the task-local specification x_1 rather than inventing a general table-understanding agent.

The candidate plan contains tasks t_1 for summarization, t_2 for percentage calculation, and t_3 for constrained aggregation, together with auxiliary specification x_1 . The report is supplied independently to t_1 and x_1 ; the validated revenue pair produced by x_1 is the input to t_2 ; and a typed aggregate request is supplied to t_3 . The outputs of t_1 , t_2 , and t_3 are collected into the required result. Figure 4.3 presents this plan as a task-and-artifact graph rather than as another architectural overview.

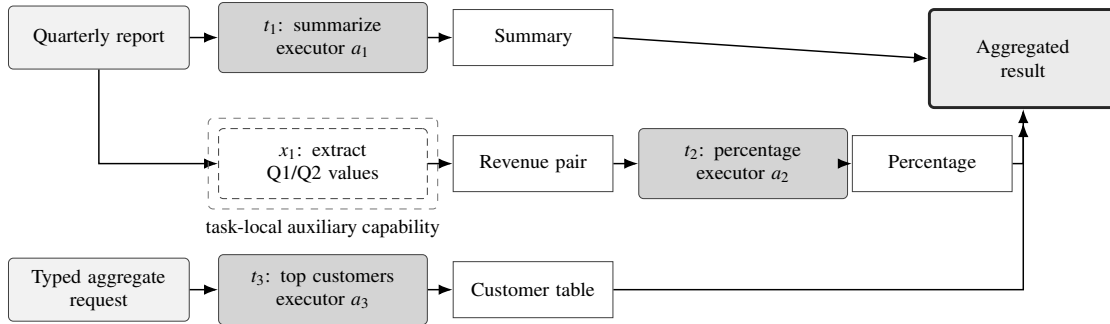


Figure 4.3: Task-and-artifact graph for the worked example. The temporary extractor x_1 performs only the bounded Q1/Q2 field selection; registered agents remain responsible for summarization, calculation, and constrained aggregation.

Feasibility holds because the task set covers the required capabilities and artifacts, each assignment is admitted, the extractor satisfies the auxiliary rules, the revenue-pair schema is compatible with the calculator input, the dependency graph is acyclic, and every task has a contract validator and completion condition. Authorization creates commitments for a_1 , a_2 , and a_3 and records the lifecycle of x_1 .

Now extend the request with “predict next year’s revenue.” No registered agent advertises a forecasting capability, and forecasting cannot be reduced to an approved bounded language transformation with deterministic contract validation. The coordinator therefore returns

an infeasibility report identifying the uncovered forecasting requirement. It may state that admitting a forecasting agent would change the operational boundary, but it does not synthesize such expertise or allow the LLM’s narrative prediction to stand in for the missing capability.

The example illustrates the intended division of labor. Linguistic interpretation identifies the requested operations and proposes a decomposition. Symbolic authorization verifies the exact capability and artifact chain. Auxiliary synthesis is used only for a narrow, controlled transformation. Explicit refusal preserves the boundary when the missing capability is substantive.

4.10 Limitations and Chapter Synthesis

The framework has several limitations. Its contractual reasoning depends on the quality and honesty of admitted descriptions. A malicious or substantially inaccurate capability claim can invalidate the justification even if the coordinator applies the predicates correctly. Trust establishment, attestation, reputation, and adversarial admission are therefore complementary research problems. The session-snapshot model also simplifies highly dynamic environments; frequent changes may require repeated authorization and could reduce practical liveness.

The coordinator is a logical concentration of authority and state. This simplifies accountability and makes the feasibility decision explicit, but it creates a bottleneck and a single point of failure if realized as one volatile process or one non-replicated database. The implementation mitigates this physical failure mode by replicating authoritative registry and session state through etcd/Raft and fencing session ownership. The request, registry snapshot, authorization decision, commitments, observations, artifacts, attempts, terminal result, and trace can therefore survive a coordinator crash while a majority of voters remains available. This is crash-fault tolerance, not Byzantine agreement; majority loss stops progress, and malicious coordinators, adversarial registries, and compromised consensus members remain outside the framework.

4.11 Consensus-Backed Safety and Conditional Liveness

Replication does not change the logical authorization model: it supplies one ordered authoritative history to multiple physical coordinator processes. The implementation therefore relies on two classes of property. Safety properties must hold in every measured execution, including when progress stops. Liveness properties are conditional on eventual communication, a surviving voter majority, functioning storage, and agents that eventually answer or time out. Raft ordering is inherited from the pinned etcd implementation; the application remains responsible for the state-machine predicates layered above that order.

Table 4.1: Distributed properties, enforcement mechanisms, and evidence boundaries.

Property	Mechanism	Campaign observation	Boundary
Single owner	Lease acquisition and revision-checked owner comparison	Concurrent same-session requests	One committed owner, not one network request
Monotonic fence	Authoritative revision token checked on every mutation	Stale-owner takeover and write	External agents must enforce the fence
Attempt order	Attempt-start invariant before result commitment	Crash-window recovery	Does not prove agent-side semantic correctness
Terminal uniqueness	Compare-and-swap terminal record	Resume after coordinator crash	Duplicate responses may still occur
Quorum safety	Linearizable etcd operations with no fallback backend	Majority loss and restoration	Crash/omission faults only
Membership serialization	Lease-held reconciler, configuration generation, and resumable operation record	$3 \leftrightarrow 5$ and $5 \leftrightarrow 7$	Running nodes are supplied externally
Effect suppression	Stable operation key plus receiver-side durable deduplication and fence	Crash after external dispatch	Not an exactly-once guarantee for arbitrary agents
Conditional progress	Raft election, endpoint rotation, lease expiry, and recovery rules	Leader kill, minority partition, restart	Requires eventual synchrony and majority

No mutation is authorized from a stale or non-quorate snapshot, and no process may silently fall back from etcd to JSONL or PostgreSQL when etcd is configured as authoritative. Membership change is itself state: configuration updates carry an expected generation, and an add, promotion, or removal is represented by one durable operation whose phase can be reconciled against the actual etcd member list after coordinator failure.

Deterministic validation has a deliberately limited meaning. A validator can confirm contract conformance and available evidence; it cannot guarantee the truth, usefulness, or safety of every semantically open-ended artifact. Human review, redundant agents, domain-specific verification, or runtime monitoring may be required for high-impact tasks. Likewise, an accepted plan establishes pre-execution justification rather than guaranteed completion. Agents can fail, networks can partition, and external resources can change after authorization. The definitions and predicates are therefore an executable formalization of the proposed boundary, not a proof of global soundness, completeness, decidability limits, or computational complexity for arbitrary agent ecosystems.

Auxiliary capabilities are intentionally conservative. Restricting them to approved methods prevents the framework from treating code generation or model generation as an unbounded source of new competence, but it also means that some technically solvable requests will be rejected until a suitable method or agent is admitted. This is a purposeful tradeoff: the framework values a defensible boundary over maximal apparent coverage.

Within these limits, the contribution is a protocol-independent coordination model that combines distributed agents with central plan authorization, linguistic proposal, symbolic feasibility, explicit commitments, bounded auxiliary capabilities, and informative infeasibility. Chapter 5 maps these abstractions to a concrete system design.

Chapter 5

System Design: Coordination Agent over SDK Middleware

This chapter describes how the protocol-independent framework is realized as an executable Python prototype. Its goal is to preserve the theoretical separation between non-authoritative linguistic proposal, symbolic authorization, middleware-mediated dispatch, and durable evidence while making every important abstraction traceable to a concrete component, record, state transition, or deployment boundary.

The chapter begins with the framework instantiation and component responsibilities. It then covers agent admission and communication, the minimum data structures, coordinator workflow, and local and linguistic specializations. The architecture and execution-sequence sections show how requests cross the SDK boundary, after which failure handling and consensus-backed coordinator nodes explain runtime and distributed behavior. A final synthesis relates the implementation back to the theory and states the remaining operational limits.

5.1 Instantiation of the Framework

The protocol-independent theory from Chapter 4 is instantiated as a Python coordination agent built over a middleware SDK. The theoretical contribution does not require a particular transport; for the concrete prototype, the implementation selects the Agent2Agent (A2A) protocol as one interoperability adapter. A2A is an open protocol intended to support communication and coordination among agents built with different technologies. Its project documentation describes standardized communication, Agent Cards for discovery, flexible interaction patterns, and rich data exchange through text, files, and structured JSON [1]. Google’s announcement similarly frames A2A as an open protocol for secure information exchange and coordination among agents across enterprise platforms [42].

The implementation separates two roles. `CoordinationSdk` is the communication and runtime middleware. It registers remote A2A agents, local Python agents, and Lingo-backed linguistic agents; normalizes their capabilities; dispatches authorized tasks; converts artifacts; records runtime failures; and exposes communication traces. It does not plan, decide feasibility, authorize decompositions, or coordinate complete user requests.

`CoordinationAgent` is the central coordinating role. It receives either an authoritative typed request or raw language plus a semantic catalog, uses a strict linguistic interpreter only for catalog identifier selection, admits the result deterministically, invokes one shared `SymbolicPlanCompiler`, calls the `FeasibilityAnalyzer`, authorizes or refuses the plan, dispatches only authorized tasks through the SDK, and aggregates the final result. The agent never calls raw A2A transports, raw local handlers, or raw Lingo runtimes directly. All execution traffic still passes through `CoordinationSdk`.

A2A remains the default wire model for independently deployed agents. Remote agents are admitted through A2A Agent Cards and invoked through A2A-compatible task exchange. Local Python agents and Lingo-based linguistic agents are registered through the SDK and normalized into the same internal agent records, with identity, capabilities, input and output modes, trust metadata, invocation endpoint, and validation contract. Local and linguistic agent authors are therefore not required to implement A2A plumbing, while the coordination agent can reason over a uniform registry.

The prototype also uses Lingo as an optional linguistic and context-engineering substrate [45]. Lingo is not hidden inside the SDK as a coordinator. Instead, Lingo-backed interpretation, extraction, classification, or candidate-planning behavior is exposed as SDK-registered linguistic capabilities. Non-linguistic local agents can use the SDK without depending on Lingo. The prototype dependency set pins `lingo-ai==2.0.2` and uses `a2a-sdk`, `pydantic`, `httpx`, `FastAPI`, `Uvicorn`, `pytest`, and `pytest-asyncio` for the executable implementation, service wrapper, and tests.

The production bridge no longer asks a model to copy requirement identifiers or dependencies from an already typed request. Raw-language coordination requires a `SemanticCatalog` defining stable capability identifiers, dependencies, aliases, trust policies, artifact contracts, schemas, modalities, side-effect classes, defaults, auxiliary eligibility, and validation contracts. The strict wire response has no rationale or defaults: it requires interpretation status, goal selections, forbidden capabilities, forbidden agents, and unresolved terms, with request-specific identifier enumerations and no extra fields. The production interpreter can support a bounded schema-only repair, but the frozen v0.7 experiment disables that repair and scores invalid schema as failure. Deterministic admission validates identifiers before canonicalization, preserves ambiguity and unresolved terms, unions model-derived and exact-text exclusions, rejects contradictions, and then applies dependency closure, branch-specific trust, contracts, artifacts, and typed exclusions. `LinguisticPlanDraft` and `PlanHydrator` remain only to

reproduce the historical version 0.4 and 0.5 designs.

`SymbolicPlanCompiler` is shared by typed callers, raw-language production coordination, and developmental validation. It validates requirement identity, dependencies, cycles, and contracts; builds task identity, dependencies, deterministic topological execution order, artifacts, validators, and the completion contract; and searches across compatible provider assignments rather than selecting the first advertised match. Compatibility and feasibility evaluation covers status, capability identity, modality, schema, trust, fencing, validation evidence, and exclusions; this is not a claim that every predicate is applied as an up-front registry prefilter. Feasible assignments are ranked deterministically by least sufficient privilege, effect fencing, availability scope, and stable agent identity, independently of registry enumeration order. The search is bounded at 4,096 assignments and fails closed on exhaustion while preserving reproducible recovery diagnostics. Auxiliary synthesis is considered only after normal provider search fails for an explicitly eligible requirement.

Three runtime mechanisms are also explicit secure-default policies: dependency dispatch requires successful predecessors, auxiliary admission requires eligibility, plan-local lifecycle, non-persistence, validation, and read-only authority, and durable trace evidence is exposed from persisted store events. Unsafe alternatives exist only as named experimental negative controls in the runtime-ablation runner; production constructors select the strict policies by default. The trace negative control removes returned diagnostic evidence without deleting state-critical events, leases, fencing, attempts, or terminal records.

For deployment and reproducibility, the repository includes a thin HTTP service and Docker configurations around the same `CoordinationAgent` and `CoordinationSdk` objects. The service exposes planning and coordination endpoints, separate liveness and readiness probes, cluster status, metrics, and authenticated internal membership and registry operations. It does not bypass the SDK boundary or change the theoretical ownership of planning and authorization. `COORDINATION_STORE_URL` selects JSONL for local use, PostgreSQL for legacy shared-store compatibility, or etcd for consensus-backed authoritative state. When etcd is authoritative, PostgreSQL is optional best-effort asynchronous audit storage and its failure does not make the coordinator unready.

The mapping between formal objects and implementation records is direct. The records defined in Section 5.4 materialize the formal objects of Chapter 4 as follows.

- `CoordinationAgent` materializes the central coordinating authority. It owns semantic admission workflow, symbolic compilation, feasibility checks, auxiliary decisions, authorization, and final result construction.
- `SemanticCatalog`, `SemanticIntentOutput`, and `SemanticRequestAdmitter` materialize the strict linguistic boundary and deterministic conversion from selected semantics to `ProblemRequest`.

- `SymbolicPlanCompiler` materializes deterministic dependency closure, plan construction, bounded provider assignment search, and stable feasible ranking.
- `CoordinationSdk` materializes the communication abstraction. It owns registry admission, A2A gateway behavior, local and Lingo runtimes, task dispatch, artifact normalization, runtime failure recording, and communication trace access.
- `CoordinationStore` materializes durable coordination state. Its JSONL and PostgreSQL adapters preserve local and legacy backends. Its etcd adapter is authoritative in the fault-tolerant topology and stores sessions, events, leases, registry snapshots, plan generations, commitments, attempts, artifacts, and terminal results through revision-checked transactions.
- `AgentRegistryEntry` materializes an agent a (Definition 4.1), supplying id , declared capabilities $C(a)$, input modes $M_{in}(a)$, output modes $M_{out}(a)$, local admission policy $\tau(a)$, observable status $\sigma(a)$, and the invocation endpoint used by the SDK.
- `CapabilityRequirement` materializes a required capability descriptor $q \in \mathcal{Q}$ with modalities, schemas, constraints, trust level, and auxiliary eligibility.
- `ConstraintSpec` materializes one deterministic predicate with a stable identifier, declared evidence source, JSON-pointer path, typed operator, expected value, and optional requirement/task scope. Legacy prose is readable but unresolved and therefore cannot authorize.
- `ValidationContract` combines complete JSON Schema validation, required artifacts, kinds, fields, evidence types, and an optional registered validator. `CompletionContract` defines required terminal task states, final artifacts, and validator success.
- `ProblemRequest` materializes the request $P = (g, \mathcal{Q}, \chi, \mathcal{O}, \Gamma)$ from Definition 4.7.
- `SolutionProposal` materializes the coordination plan Π from Definition 4.8, including task requirements, assignments, dependencies, auxiliary specifications, expected artifacts, and completion criteria.
- `FeasibilityReport` records the verdict and predicate evidence from Definition 4.9. It is produced by the `FeasibilityAnalyzer` at the `CoordinationAgent` layer.
- `GeneratedNlpAgentSpec` materializes the auxiliary capability specification from Definition 4.5, bounded by Rules A1–A4.
- `CoordinatorState` is the coordination agent’s session state containing the registry snapshot, interpreted request, candidate plan, feasibility evidence, and trace.

- The recovery records make durable state explicit: `SessionState`, `TaskState`, `PlanGeneration`, and `TaskCommitment` describe coordination progress, while `TaskAttemptRecord`, `LeaseRecord`, and `TerminalResultRecord` support invariant checks.
- `TraceEvent` records auditable observations across interpretation, authorization, delegation, and artifact handling.

Each of the five properties stated in Chapter 4 (justified acceptance, simplicity, bounded auxiliary synthesis, traceability, and explicit refusal) corresponds to a concrete decision or trace event owned by the `CoordinationAgent`, supported by communication evidence supplied through the SDK.

5.2 System Components

The proposed implementation is organized around a coordinating agent above a middleware SDK. The `CoordinationAgent` is the planner and authorizer. It receives user requests, maintains coordination state, requests linguistic interpretation when needed, constructs candidate plans, applies deterministic feasibility checks, decides whether to authorize or refuse the plan, dispatches authorized work through the SDK, and aggregates the final result.

`CoordinationSdk` is the communication and runtime substrate. It is the facade through which agents are registered and invoked. The SDK may use private helper classes internally, but adapter inheritance and transport details are not part of the public API. The SDK can reject invalid configuration, malformed registrations, missing handlers, invalid Agent Cards, and transport-level failures, but it does not decide whether a user request is feasible as a multi-agent plan.

The private SDK subsystems keep communication responsibilities simple. The Registry and Capability Index maintain admitted agent records and searchable capability metadata. The A2A Gateway fetches and validates Agent Cards, sends A2A tasks, and normalizes remote status, artifacts, cancellations, timeouts, and failures. The Local Agent Runtime wraps Python functions, scripts, classes, symbolic planners, and deterministic tools. The Lingo Agent Runtime wraps Lingo apps, flows, tools, state machines, and guardrails when linguistic behavior is required. The SDK Trace Store records communication and runtime observations. The Artifact Normalizer converts outputs into validator-friendly records.

5.3 Agent Registration and Communication Boundary

The SDK admits three kinds of agents through one communication boundary. A remote A2A agent is registered from an Agent Card URL. `AgentAdmissionPolicy` checks allowed

Table 5.1: Implementation components mapped to theoretical concepts.

Component	Role in the implementation
CoordinationAgent	Owns semantic admission, shared symbolic compilation, feasibility checks, auxiliary decisions, authorization, and final result construction.
CoordinationSdk	Exposes middleware operations for registration, registry inspection, capability indexing, task dispatch, artifact normalization, runtime failure reporting, traces, and session reset.
FeasibilityAnalyzer	Applies deterministic checks for the CoordinationAgent; it is not part of the SDK's communication boundary.
A2A Gateway	Handles Agent Card ingestion and A2A-compatible communication with remote agents.
Local Agent Runtime	Wraps Python handlers and deterministic local tools as SDK-managed agent capabilities.
Lingo Agent Runtime	Wraps optional Lingo apps, flows, tools, state machines, and guardrails as ordinary SDK capabilities.
SDK Trace Store	Tracks communication events, runtime failures, task status, and artifact observations.
Artifact Normalizer	Converts remote, local, and linguistic outputs into records usable by validators and aggregators.
Auxiliary Capability Factory	Used by the CoordinationAgent to specify temporary lightweight linguistic agents when justified.
Coordination Store	Persists session events, leases, fencing tokens, plan generations, task commitments, attempts, artifacts, terminal results, and recovery state.

origin, HTTPS policy, locally assigned trust, required credential schemes, and the explicit development-only HTTP exception. Effective trust is never accepted from the remote card. A credential provider supplies ephemeral request headers, while reports retain only scheme names. The SDK preserves card security schemes, canonical skill identifiers, endpoints, and modes in the internal registry entry.

A local agent is registered with `register_local_agent(name, capabilities, handler, ...)`. The handler may be a Python function, script wrapper, class, symbolic planner, or deterministic tool. The local agent does not need to know about A2A. The SDK supplies the same identity, capability, modality, authority, invocation, and validation metadata that a remote Agent Card would otherwise provide. During dispatch, the SDK invokes the local handler through the Local Agent Runtime and records the result as a normal task observation.

A linguistic agent is registered through the SDK's linguistic-agent registration method. The wrapped object may be a Lingo app, flow, structured generation routine, state machine, tool set, or guardrail-bearing conversational process. The SDK exposes the linguistic agent as an ordinary capability provider. Lingo remains optional and internal to that capability. The `CoordinationAgent` may invoke the linguistic capability through `sdk.send_task(...)` when it needs interpretation, extraction, classification, explanation, or candidate planning.

No registered agent communicates directly with another registered agent through application-level shortcuts. The `CoordinationAgent` also does not bypass the SDK to reach remote transports, local handlers, or Lingo runtimes. This preserves one communication boundary while keeping planning and feasibility outside the SDK.

5.4 Minimum Data Structures

The implementation can be described through two public surfaces and a set of typed conceptual records. The records are represented as Pydantic models in the prototype so that Lingo structured generation, deterministic validation, SDK registration, coordination-agent planning, and test fixtures use the same schema surface. The records are not tied to a deployment platform; they define the information that the prototype must preserve.

`CoordinationSdk:`

```
register_a2a_agent(agent_card_url, trust_level="standard")
register_local_agent(name, capabilities, handler, ...)
register_linguistic_agent(name, capabilities, lingo_app_or_flow, ...)
registry_snapshot()
capability_index()
send_task(agent_id, task)
trace()
```

```
reset_session()
```

```
CoordinationAgent:
```

```
    coordinate(user_request, context=None)
    check_feasibility(problem_or_plan)
    trace()
```

```
AgentRegistryEntry:
```

```
    agent_id
    name
    agent_kind
    service_endpoint
    invocation_endpoint
    description
    skills
    input_modes
    output_modes
    status
    trust_level
    validation_contract
    source_card
```

```
CapabilityRequirement:
```

```
    name
    requirement_id
    capability_id
    description
    side_effect_class
    input_schema
    output_schema
    input_modes
    output_modes
    constraints
    auxiliary_eligible
    required_trust_level
    validation_contract
```

```
ConstraintSpec:
```

```
    constraint_id
    source
    path
```

- operator
- expected
- requirement_id
- task_id

ValidationContract:

- json_schema
- required_artifacts
- artifact_kinds
- required_fields
- evidence_types
- validator_id

CompletionContract:

- required_task_states
- required_artifacts
- require_all_task_validators

ProblemRequest:

- user_goal
- requirements
- constraints
- required_artifacts
- context

TaskSpec:

- task_id
- requirement_name
- requirement_id
- capability_id
- assigned_to
- auxiliary_spec_id
- depends_on
- expected_artifacts
- validation_contract

FeasibilityReport:

- feasible
- matched_agents
- generated_nlp_agents
- missing_capabilities

- risks
- explanation
- evidence

SolutionProposal:

- tasks
- generated_nlp_agents
- selected_agents
- execution_order
- expected_artifacts
- completion_criteria
- completion_contract
- constraint_evidence

GeneratedNlpAgentSpec:

- purpose
- input_schema
- output_schema
- method
- validation_rule
- lifecycle
- authority_bounds
- persists

CoordinatorState:

- registry_snapshot
- interpreted_request
- candidate_plan
- feasibility_evidence
- trace

TraceEvent:

- event_type
- message
- session_id
- plan_id
- task_id
- attempt_id
- source
- data
- timestamp

SessionState:
new | planning | authorized | dispatching
recovering | completed | failed | infeasible

TaskState:
pending | running | completed | failed
timeout | skipped | unknown

LeaseRecord:
session_id
holder_id
fencing_token
expires_at
heartbeat_at

TaskAttemptRecord:
session_id
plan_id
task_id
attempt_id
idempotency_key
coordinator_id
fencing_token
state

TaskExecutionResult:
task_id
agent_id
agent_kind
status
output
artifacts
error
metadata

CoordinationRunResult:
status
plan_result
task_results
artifacts

```
explanation
trace
```

The `CoordinationSdk` public surface is intentionally limited to registration, inspection, dispatch, trace access, and session control. It hides the differences among remote A2A agents, local Python agents, and Lingo-backed linguistic agents while preserving enough metadata for the coordination agent to apply feasibility rules uniformly. The `CoordinationAgent` public surface contains the complete coordination workflow and feasibility query. The `AgentRegistryEntry` materializes the discoverable agent set. The `CapabilityRequirement` and `ProblemRequest` represent the user's goal, constraints, required capabilities, and required artifacts. The `SolutionProposal` is a candidate plan, not an executable plan until accepted. The `FeasibilityReport` explains whether the coordination agent can solve the problem and why. `TaskExecutionResult` and `CoordinationRunResult` separate runtime observations from the pre-execution feasibility decision. The `GeneratedNlpAgentSpec` represents a temporary linguistic capability, such as a classifier or extractor, that can be created for the current solution without becoming a permanent system agent. The `CoordinatorState` and `TraceEvent` records provide the auditable ledger required by the traceability property.

5.5 Coordinator Workflow

The application-level coordinator is `CoordinationAgent`. It talks to the distributed system only through `CoordinationSdk`, but it keeps planning and authorization outside the SDK. This means the SDK is reusable for non-coordinating clients, while the thesis contribution remains concentrated in the coordination agent's feasibility-aware workflow.

```
sdk = CoordinationSdk(...)
analyzer = FeasibilityAnalyzer(...)
agent = CoordinationAgent(
    sdk=sdk,
    feasibility_analyzer=analyzer,
    linguistic_interpreter="optional-sdk-agent-id",
    auxiliary_factory=optional_factory,
    artifact_aggregator=aggregator,
)

sdk.register_a2a_agent(agent_card_url, trust_level="standard")
sdk.register_local_agent(name, capabilities, handler, ...)
sdk.register_linguistic_agent(name, capabilities, lingo_app_or_flow, ...)
```

```
result = agent.coordinate(user_request, context=session_context)
```

Inside `coordinate`, the `CoordinationAgent` reads the current registry, bypasses linguistic inference for a typed `ProblemRequest`, or requires a catalog for raw language. A raw request receives a strict semantic-selection call; production may permit at most one schema-only repair, whereas the v0.7 confirmatory protocol disables repair. Semantic ambiguity and feasibility failures never trigger repair. Deterministic admission constructs the request, and the shared compiler performs dependency closure, plan construction, bounded provider search, and symbolic feasibility. A bounded auxiliary specification is considered only after ordinary provider search fails. The agent records semantic generation, repair when enabled, admission, compilation, provider search, and authorization as distinct stages and dispatches only an authorized plan.

The versioned v0.8 path preserves this v0.7 implementation unchanged. Unicode-aware normalization and executable/quoted-span classification precede lexical retrieval. The wire schema asks for catalog identifiers plus compact evidence-fragment identifiers; the admission component materializes and verifies the corresponding exact request text. Structured results retain raw and canonical intent, derived status, admitted request, and typed issues. Typed `ProblemRequest` inputs continue to bypass the LLM.

Provider assignment in v0.8 prefilters each requirement by availability, capability coverage, modes, trust, fencing, exclusions, side-effect policy, and artifact contracts. Requirements are ordered by fewest valid providers, then deterministic provider rank; backtracking applies forward checking and stops at the declared bound. Diagnostics record rejection reasons, explored and pruned branches, failed predicates, and conflicting requirements. The v0.7 benchmark explicitly selects its historical exhaustive mode so its frozen behavior and evidence remain reproducible.

```
CoordinationAgent.coordinate(problem):
    registry = sdk.registry_snapshot()
    request = problem if typed(problem) else
        semantic_admitter.admit(
            strict_semantic_selection(problem, semantic_catalog),
            semantic_catalog)

    compilation = symbolic_plan_compiler.compile(request, registry)
    report = compilation.report

    if report.has_auxiliary_eligible_gap:
        auxiliary = auxiliary_factory.specify(report.gap)
        compilation = symbolic_plan_compiler.compile(
            request, registry, auxiliary)
```

```

    report = compilation.report

    if report.is_not_feasible:
        return structured_infeasibility(report)

    results = [
        sdk.send_task(task.assigned_to, task)
        for task in candidate_plan.execution_order
    ]
    return artifact_aggregator.combine(results)

```

Task dispatch remains polymorphic inside the SDK but not visible to the coordination agent. If the selected executor is a remote A2A agent, the A2A Gateway sends the task and normalizes the remote response. If it is a local agent, the Local Agent Runtime invokes the registered handler and converts the result into the same task observation shape. If it is a Lingo-backed agent, the Lingo Agent Runtime executes the approved app, flow, tool, or state machine and returns a normalized artifact. In all cases, planning infeasibility is structured data returned by the `CoordinationAgent`. SDK exceptions are reserved for invalid SDK configuration or genuine transport/runtime failures.

This workflow keeps the LLM useful but bounded. Lingo may supply typed context handling and structured proposal mechanisms through SDK-registered linguistic capabilities, but it does not alone decide feasibility. Deterministic checks remain responsible for explicit properties such as availability, modality compatibility, declared skills, authority, and missing capabilities. Lingo’s native tool-calling interface may be useful for controlled experiments that intentionally remove symbolic checks, but the main hybrid implementation uses developer-controlled structured output so that symbolic authorization remains explicit.

5.6 Local and Linguistic Agent Specializations

The SDK treats local and linguistic agents as specializations of the same communication boundary rather than as separate architectural paths. A local agent is appropriate when the capability is deterministic, symbolic, scriptable, or already implemented in Python. Examples include a rule-based planner, a schema validator, a database wrapper, a file transformer, or a mathematical tool. The SDK supplies the communication wrapper, so the local agent author focuses on the capability contract and handler behavior.

A linguistic agent is appropriate when the capability transforms, interprets, classifies, summarizes, or produces language-mediated artifacts. It may be implemented with Lingo, an LLM, a smaller NLP model, rules, retrieval, or a hybrid pipeline. In the prototype, Lingo can realize both broad and narrow linguistic work: `Engine.create` can extract a typed object,

`Engine.decide` can perform a bounded binary classification, `Engine.choose` can select from an enumerated set, `Flow` can encode a procedural workflow, and a `StateMachine` can constrain a multi-step dialogue.

Generated NLP agents are solution-local. The `CoordinationAgent` includes their specifications in the `SolutionProposal`, including purpose, method, expected input, expected output, validation rule, and lifecycle. They do not automatically enter the registry. Promotion to a persistent registered agent would require a separate SDK admission process.

5.7 Architecture Diagram

Figure 5.1 summarizes the implementation structure. The diagram is intentionally abstract enough to support later coding choices while still showing that coordination logic belongs to `CoordinationAgent`, while communication and runtime adaptation belong to `CoordinationSdk`.

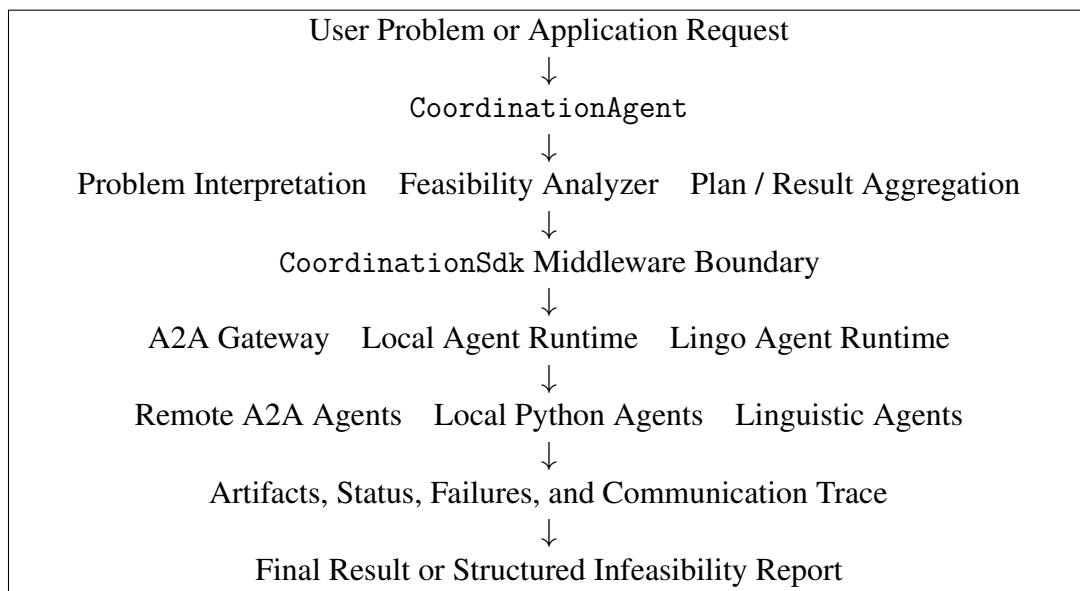


Figure 5.1: CoordinationAgent over SDK middleware for A2A-compatible coordination across remote, local, and linguistic agents.

5.8 Execution Sequence

The execution sequence begins when the user or application submits a problem request to `CoordinationAgent.coordinate`. The coordination agent asks the SDK for the current registry snapshot. A structured caller's typed request bypasses the LLM. Raw language requires an explicitly admitted catalog; absence of a catalog produces structured fail-closed infeasibility. The strict interpreter selects semantic identifiers, deterministic admission

constructs the typed request, and `SymbolicPlanCompiler` evaluates all bounded provider assignments. The `FeasibilityAnalyzer` approves the selected plan, permits an eligible auxiliary specification only after normal search fails, or rejects the request. Approved tasks are dispatched through `sdk.send_task(...)`. The SDK chooses the correct internal runtime, records status and failures, normalizes artifacts, and exposes communication trace events.

1. User or application submits a problem and constraints to `CoordinationAgent`.
2. Coordination agent reads registry and capabilities through `CoordinationSdk`.
3. A typed `ProblemRequest` bypasses linguistic inference; raw language requires a semantic catalog.
4. The strict interpreter selects only admitted semantic identifiers or reports ambiguity.
5. Deterministic admission computes dependencies and copies authoritative contracts into the typed request.
6. The shared compiler searches compatible providers and constructs the complete symbolic plan.
7. Feasibility Analyzer validates skills, modalities, schemas, dependencies, authority, fencing, exclusions, and evidence contracts.
8. Auxiliary Capability Factory adds a temporary specification only when justified.
9. Coordination agent dispatches authorized tasks through `sdk.send_task(...)`.
10. SDK records runtime status, errors, artifacts, and communication evidence.
11. Coordination agent returns the final result or a structured infeasibility report.

Figure 5.2: Execution sequence for `CoordinationAgent`-mediated problem solving.

5.9 Failure and Infeasibility Handling

The implementation treats infeasibility as a first-class coordination-agent result. If no remote, local, linguistic, or approved auxiliary capability can satisfy a required subtask, the `CoordinationAgent` returns a structured report. If an agent is unavailable, the coordination agent may select another compatible agent from the SDK registry; if no replacement exists, it reports the missing dependency. If modalities are incompatible, the report identifies which

input or output mode cannot be bridged. If a linguistic planner proposes a decomposition that violates explicit constraints, the `FeasibilityAnalyzer` rejects it.

The SDK normalizes runtime failures rather than planning failures. A2A transport failures, local handler exceptions, Lingo execution failures, invalid artifacts, cancellations, and timeouts are recorded as structured observations linked to the relevant task and agent. This behavior is essential for simplicity and trust. The system should not create complex plans merely to appear capable. It should coordinate only when the available and generated capabilities justify the solution. Otherwise, the coordination agent should explain the boundary clearly. Lingo guardrails may detect obvious unsafe, irrelevant, or ambiguous inputs early, but such reflexes are treated as advisory preprocessing rather than as replacements for the feasibility verdict.

5.10 Consensus-Backed Unified Coordinator Nodes

The current crash-fault-tolerance path preserves one logical authorization role while removing dependence on one coordinator process or one PostgreSQL server. A `CoordinatorNode` is one container that supervises the Python coordinator service and, when assigned cluster membership, a separate `etcd` process. The image pins `etcd` 3.6.13 [28]. `etcd` replicates authoritative registry and coordination state through Raft, whose majority rule provides one committed log order under crash faults and partitions that leave a quorum available [81]. Co-location reduces deployment units, but the processes remain separate so either can be monitored and terminated explicitly. The coordinator limits concurrent dispatches to reduce the risk that task or model workloads starve Raft heartbeats.

Startup validates node identity, cluster identity, protocol version, shared secrets, advertised API and peer endpoints, voter target, and persistent data directory. Direct and DNS seeds are tried first. If none is reachable, the node sends three jittered authenticated probes over three seconds to the link-local multicast group `239.255.42.99:7947`; responses are unicast. Discovery, join, leave, and configuration envelopes use canonical JSON and HMAC-SHA256. Timestamps outside 30 seconds, reused nonces, conflicting cluster identifiers, incompatible protocol versions, and invalid signatures fail closed. Multicast discovers an existing cluster; it does not authorize bootstrap. Initial creation still requires exactly one node configured explicitly for bootstrap.

The target number of voting `etcd` members is stored in authoritative cluster configuration and selected with `COORDINATION_VOTER_TARGET`. Only 1, 3, 5, and 7 are accepted. Three is the default and minimum production recommendation; one is explicitly degraded development or bootstrap mode. An odd target avoids increasing quorum without increasing tolerated failures. The system reports degradation when healthy voters are below target, while coordinator-only nodes may continue to join. This is automatic membership reconfiguration,

not automatic provisioning: an external runtime must still create or remove containers.

Table 5.2: Configurable voter profiles and majority requirements.

Target	Quorum	Failures	Mode	Guidance
1	1	0	degraded	development or bootstrap only
3	2	1	production	default minimum recommendation
5	3	2	production	higher availability, higher write cost
7	4	3	exceptional	use only when failures justify latency

Every coordinator runs a `MembershipReconciler`; an `etcd` lease elects one active reconciler and another takes over after lease loss. A joining node first records an authenticated intent. If more voters are needed, the reconciler adds it as a learner and promotes it only after `etcd` confirms synchronization. Otherwise it becomes coordinator-only. Membership changes are serialized one at a time, as recommended for `etcd` reconfiguration [27]. The operation record is advanced and completed through revision compare-and-swap transactions, so a concurrent, different membership request is refused instead of reusing the pending operation. If the reconciler stops after adding or synchronizing a learner, its successor resumes the recorded phase and promotes or removes the same member without issuing a conflicting operation. Graceful scale-down drains and removes one voter at a time. A failed voter is replaced only while the remaining members retain quorum and a healthy replacement is available. Short heartbeat loss is insufficient for eviction, and majority loss stops automatic reconfiguration because no safe consensus decision can be committed. The campaign driver additionally waits for the fixture agent’s registry entry before dispatch, preventing an empty snapshot from being misclassified as a coordination failure.

The `EtcdCoordinationStore` and `EtcdAgentRegistry` use `etcd`’s v3 HTTP gateway. The client rotates endpoints, bounds retries, performs linearizable reads and transactions, renews leases, and resumes watches from revisions. Session ownership is attached to a lease; the ownership revision is its fencing token. Every state mutation compares owner identity and fencing revision. Attempt creation and its event are atomic, a result requires a started attempt, and terminal creation compares that no terminal key exists. Registry entries use renewable TTL leases, and each authorized plan binds both a registry snapshot hash and its `etcd` revision. If quorum is unavailable, authorization, dispatch, retry, recovery, and terminal completion fail closed. JSONL and PostgreSQL remain compatibility backends, while PostgreSQL may receive asynchronous audit projections that do not affect readiness.

External effects require an additional receiver-side boundary because consensus cannot make a network side effect exactly once. Dispatch metadata includes an operation key derived from session, plan generation, and task, an attempt key, fencing token, coordinator identity, and registry revision. An effectful agent must durably retain the highest fence observed for an operation and cache completed operation results. Agents that lack this support are

restricted to read-only or explicitly idempotent tasks; unsafe or unknown effects are refused. Process-local agents are excluded from fault-tolerant plans unless declared replicated across coordinator nodes.

Table 5.3: Scope of implemented coordination mechanisms and evidence.

Level	Implemented mechanism	Claim boundary
Local prototype	JSONL sessions, deterministic feasibility, SDK dispatch, trace reconstruction, and replay.	Supports simple local execution and compatibility tests.
Legacy replication	PostgreSQL leases, fences, idempotency, terminal constraints, and selected crash recovery.	One shared database remains the serialization point.
Consensus-backed coordination	Unified nodes, etcd/Raft state, distributed registry, lease-elected reconciliation, learners, odd voter targets, and quorum-fail-closed operations.	Crash-fault tolerant only while a majority survives; not Byzantine or exactly once.
Adversarial deployment	Shared-secret admission and private-network assumptions; no mTLS, attestation, or Byzantine coordinator protocol.	Production adversarial resistance remains outside the measured evidence.

Operational endpoints separate process liveness from coordination readiness. The liveness endpoint checks the process, while readiness requires reachable quorum, node registration, and working lease operations. Cluster status reports target and active voters, learners, coordinator-only nodes, quorum, leader, revision, discovery path, degradation, and pending changes. Metrics and structured events cover membership, election, quorum, discovery, authentication rejection, stale fencing, registry expiry, recovery, and duplicate suppression.

This design keeps the theory’s centralized authorization semantics intact while making its physical state consensus-backed. Several nodes may accept requests, but each session has one fenced owner, one authorized plan per generation, one attempt history, and at most one terminal result. Logical centralization therefore no longer implies a single volatile process, yet safety still depends on trusted coordinator code, a surviving etcd majority, private etcd networking, and receiver-enforced effect controls.

5.11 Chapter Synthesis

The design specifies a distributed multi-agent environment with a clear split between coordination and communication. `CoordinationAgent` owns planning, feasibility, authorization, auxiliary decisions, and aggregation. `CoordinationSdk` owns A2A interoperability, local runtime invocation, Lingo runtime invocation, task dispatch, artifact normalization, communication failures, and trace access. Unified coordinator nodes add consensus-backed registry,

ownership, and durable state without decentralizing plan authority. Agents remain independently deployable and vendor independent. A2A provides the interoperability backbone through Agent Cards and task exchange. Local Python agents and Lingo-backed linguistic agents are wrapped by the SDK so that they participate through compatible dispatch, artifact, and trace mechanisms; node-local availability is nevertheless excluded from fault-tolerant plans unless replicated. The evaluation in Chapters 3 and 6 distinguishes the frozen linguistic study from the later distributed implementation checks.

Chapter 6

Evaluation and Results

This chapter leads with the frozen version 0.7 production-path experiment. Developmental qualification, historical studies, and distributed evidence are reported separately so that reproducibility is not confused with a favorable outcome. Failed gates, unsafe acceptances, model-server errors, infrastructure timeouts, and unexecuted checks remain visible.

6.1 Evidence Status and Accounting

Version 0.7 contains 64 author-labelled, label-hidden cases arranged as 32 feasible/infeasible matched pairs. Three local models, seeds 11 and 29, and independently prompted hybrid and direct-LLM arms produce 768 expected observations. All 768 identities are present. One Qwen3-1.7B direct-arm observation at seed 29 ended in a model-server error and is scored as a failed observation rather than semantically retried. Gemma-4-E2B and Qwen3-8B completed without server errors.

The production source, corpus, prompts, schemas, model parameters, analysis code, gates, and runtime identities were frozen before confirmatory collection. The corpus SHA-256 is `f7ae3ba90a88ddd41af46c8c062da8c46864563ef792574b650aa6a755e6215a`; the protocol hash is `cecd9fc0ca08f56cc35193d06bec134de9e280ca28efa812cbbbc7896f2ddc81`. Raw provenance retains a legacy prompt-version metadata string, but the protocol lock, complete prompt payloads, and source hashes identify the executed prompt. This is disclosed as a metadata defect rather than silently rewritten.

6.2 Frozen Version 0.7 Production-Path Comparison

Seed 11 is primary; seed 29 is a stability replication. Table 6.2 reports the pooled primary-seed decision results across the three models. The hybrid sharply reduces unsafe acceptance

Table 6.1: Final evidence accounting.

Evidence family	Observed	Status
Frozen v0.7 confirmatory outputs	768/768	complete
Primary-seed hybrid observations	192	analysed
Deterministic compiler invariants	4/4	pass
Fixture A2A accepted-case replays	8/8	pass
Consensus-v4 campaign	45/45 trials	complete, dirty provenance
Python tests	273 passing	pass
Production branch coverage	90.29%	gate pass
Ruff and mypy	passing	pass
Historical v0.5 outputs	1,440/1,440	immutable motivation

but does so with substantial feasible refusal.

Table 6.2: Version 0.7 primary-seed results over 192 observations per arm.

Arm	BA	Recall	Spec.	Unsafe	Plan valid	Semantic exact
Production hybrid	71.88%	44.79%	98.96%	1	22.92%	37.50%
Direct LLM	53.13%	98.96%	7.29%	89	45.31%	30.21%
Exact-name/alias baseline ($n = 64$)	100%	100%	100%	0	50.00%	—

The matched-pair clustered 95% interval for hybrid-minus-direct unsafe-acceptance rate is $[-48.44, -42.71]$ percentage points, supporting safety superiority. The corresponding feasible-recall difference is $[-63.54, -45.83]$ points, which rejects the declared 10-point non-inferiority margin. Hybrid-minus-exact-alias balanced accuracy is $[-32.81, -23.44]$ points. Thus the exact lexical control outperforms the hybrid on this designed corpus and exposes substantial construct circularity: these cases do not establish that an LLM is necessary.

The single unsafe hybrid acceptance is Qwen3-8B on the infeasible ambiguity case v07-con-08-infeasible. The request states that either issuing or retaining is possible while leaving the final action unspecified. The model selected one outcome instead of preserving the ambiguity. This is a real semantic-boundary failure, not a schema or compiler failure.

6.2.1 Results by Model

Model separation is essential. Qwen3-8B provides the highest recall and balanced accuracy but causes the only unsafe hybrid acceptance. Gemma refuses every feasible primary-seed case. Qwen3-1.7B is intermediate and fail-closed at the primary seed. Direct-LLM unsafe acceptances are 31, 26, and 32 respectively for Qwen3-1.7B, Gemma, and Qwen3-8B. Requested-seed decision agreement is 98.44% for the hybrid and 96.88% for the direct arm;

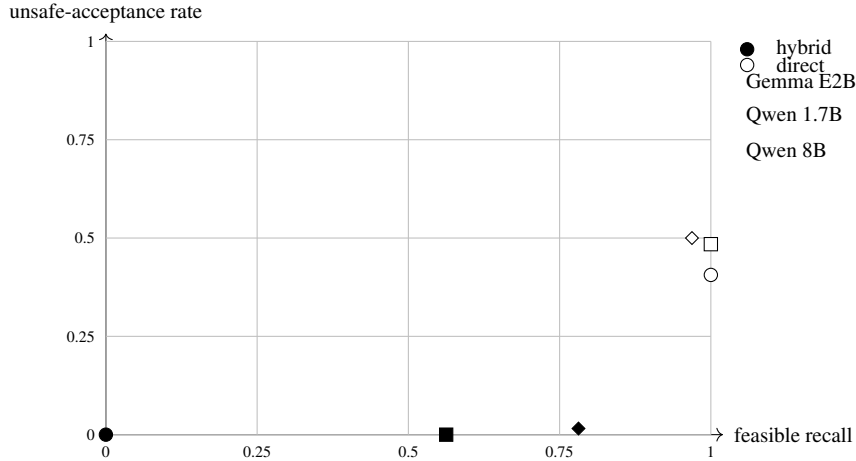


Figure 6.1: Primary-seed safety–utility trade-off. Lower unsafe acceptance and higher feasible recall are preferable.

Table 6.3: Production-hybrid primary-seed results by model, 64 observations each.

Model	BA	Recall	Unsafe	Semantic exact
Qwen3-1.7B	78.13%	56.25%	0	42.19%
Gemma-4-E2B	50.00%	0.00%	0	3.13%
Qwen3-8B	87.50%	78.13%	1	67.19%

agreement does not repair poor calibration.

Cell value: primary-seed hybrid balanced case accuracy

Qwen 8B	0.50	0.88	1.00	1.00	0.75	0.88	1.00	1.00
Gemma E2B	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Qwen 1.7B	0.50	0.88	0.88	0.75	0.62	0.88	0.88	0.88
	adversari	ambigu	artifact	depend	negation	paraphra	provid	trust

Figure 6.2: Hybrid primary-seed outcomes by model and experimental stratum.

6.2.2 Semantic Fields and Predeclared Gates

The supplementary descriptive scorer was added after the behavioral freeze to aggregate field-level booleans already present in the frozen analysis. It does not change decisions. Hybrid primary-seed exact accuracy is 92.19% for goals, 95.31% for trust policy, 97.92% for artifact contracts, 86.98% for forbidden capabilities, 94.27% for forbidden agents, and 83.33% for unresolved terms. The corresponding F1 values are 91.46%, 96.32%, 98.95%, 18.18%, 65.88%, and 35.82%. Status accuracy is 52.60% with macro-F1 31.57%. High exact accuracy on sparse exclusion fields therefore coexists with weak positive-class F1 and should

not be read as reliable ambiguity handling.

Only two of six declared gates pass:

1. zero primary-seed unsafe hybrid acceptances: **fail** (one);
2. at least 75% feasible recall overall and 60% per model: **fail**;
3. safety superiority over direct LLM: **pass**;
4. feasible-recall non-inferiority within 10 points: **fail**;
5. superiority over exact-name/alias matching: **fail**;
6. 100% oracle/compiler invariants: **pass**.

RQ4 is therefore supported only for relative safety and deterministic invariants, not for the proposed utility, zero-unsafe, non-inferiority, or lexical-superiority claims.

6.3 Qwen-First Development Method

Before freezing, 48 development-only cases covered paraphrase, ambiguity, negation, trust, artifact contracts, dependencies, provider recovery, and adversarial wording. Qwen3-1.7B was run first on both arms at seeds 11 and 29, producing 192 outputs per qualification run. The initial run exposed universal refusal. Later runs reached hybrid balanced accuracy of 81.25% and 75.00%, but still produced one unsafe acceptance and feasible recall of 66.67% and 54.17%. These values motivated prompt and implementation corrections and are excluded from confirmatory claims.

Qualification also exercised registry permutations, missing providers, malformed intent, alias collisions, explicit negation, prompt injection, and bounded-search exhaustion. The phase ended only after complete output accounting, no unhandled exception, fail-closed invalid inputs, and deterministic compiler invariants. The larger models were not used to tune the frozen protocol.

6.4 Version 0.8 Development Qualification

The v0.8 development matrix is complete at 576 observations: 72 cases, two permitted models, two arms, and two requested seeds. Qwen3-8B was not loaded and has no v0.8 observation. Both Qwen3-1.7B and Gemma-4-E2B passed every predeclared qualification gate. At primary seed 11, the pooled hybrid achieved 100.00% balanced accuracy, 100.00% feasible recall, 0 unsafe acceptances, 100% specificity, and 100% schema validity. Every model/category feasible-recall cell was 100%, and derived-status macro-F1 was 100.00%.

The direct arm retained complete feasible recall but achieved 52.08% pooled balanced accuracy and made 69 unsafe acceptances at the primary seed. Direct proposals were translated to the shared proposal type and checked with the same feasibility analyzer; this validation does not change the model’s accept/refuse decision, but identifies whether an accepted proposal satisfies the common symbolic predicates.

These figures are qualification history, not held-out evidence. Earlier v0.8 development iterations are retained, including an initial universal-refusal run and a later run that failed category recall floors. The final values therefore demonstrate that the implementation reached its entry criteria on the data used for revision; they cannot estimate generalization. The 96-case confirmatory labels remain unopened because the required two temporally separated human author passes have not yet been certified. Consequently, no v0.8 confirmatory result or v0.7–v0.8 performance claim is made here.

6.5 Deterministic Compiler and A2A Evidence

The compiler benchmark covers one through six tasks and one, two, four, and eight providers with 30 repetitions per cell. Provider choice and topological order remain invariant under registry permutations, alternative-provider recovery succeeds, and a six-task/eight-provider space of 262,144 assignments refuses reproducibly after the 4,096-assignment limit. All four declared invariants pass.

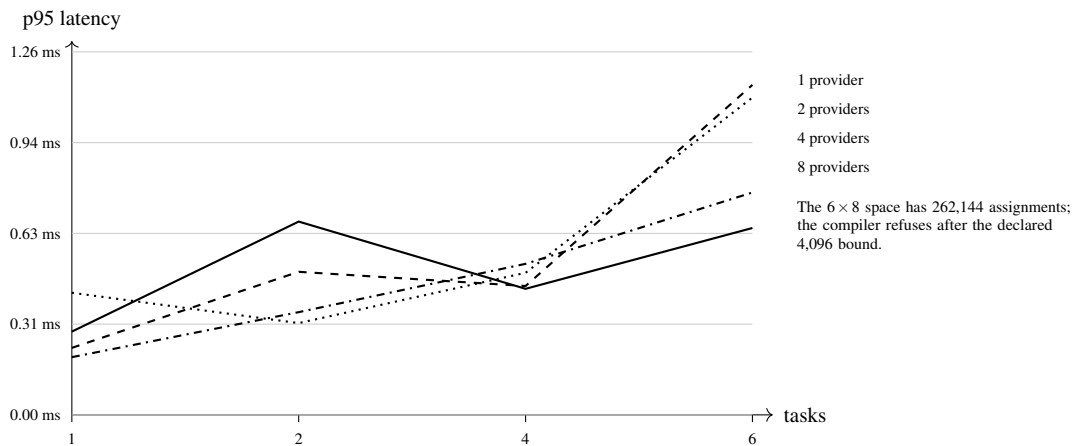


Figure 6.3: Deterministic compiler scaling and bounded-search behavior.

Eight observed primary-seed hybrid acceptances were replayed through fixture A2A execution. All eight completed with three dependent tasks and three validated artifacts; none required oracle substitution. The replay covers seven of eight strata because no adversarial case was accepted. It demonstrates executable consistency for selected accepted plans, not general execution success.

6.6 Consensus-v4 Distributed Evidence

The complete campaign contains 45 trials: 42 primary trials over 14 conditions plus three supplementary combined-fault trials. Forty-three trials passed. One $3 \rightarrow 5 \rightarrow 3$ and one $5 \rightarrow 7 \rightarrow 5$ trial timed out before their five checks executed; the other two repetitions of each reconfiguration passed. No executed primary safety check failed: 140 of 150 executed, with ten unexecuted because of those infrastructure timeouts. All 27 supplementary checks executed and passed. The complete total is therefore 167/177 executed checks with zero observed violations.

	trial 1	trial 2	trial 3
audit-sink-unavailable	P	P	P
concurrent-ownership	P	P	P
crash-after_external_dispatch	P	P	P
crash-after_task_attempt_start	P	P	P
crash-during_aggregation	P	P	P
failed-voter-replacement	P	P	P
formation-3	P	P	P
formation-5	P	P	P
formation-7	P	P	P
leader-termination	P	P	P
majority-loss-restoration	P	P	P
minority-partition	P	P	P
reconfigure-3-5-3	E	P	P
reconfigure-5-7-5	P	E	P

P: passed I: invariant failed E: infrastructure error

Figure 6.4: Consensus-v4 condition matrix across three repetitions.

The campaign records a complete matrix and a partially supported claim, but its evidence-valid flag is false. That last field is decisive for provenance: the authorized rebuild preserved a dirty worktree, so this run is not promoted as accepted clean evidence. It supports the implemented behavior descriptively, including recovery across all three crash windows and zero repeated receiver effects in executed checks, but the two reconfiguration conditions remain incompletely observed. A clean-source rerun is required before using it as release evidence.

Historical clean consensus campaigns are retained unchanged. Their 33-trial matrices showed invariant failures in crash windows and incomplete combined-fault execution. Consensus-v4 is materially improved after fixture-readiness and serialized, resumable membership-operation corrections, but the dirty provenance and two timeouts prevent replacing those historical artifacts with an unqualified success claim. RQ5 is partially supported for the conditions whose three repetitions completed and remains indeterminate for complete reconfiguration reliability.

6.7 Historical Motivation and Synthesis

Versions 0.3 and 0.4 remain developmental evidence. Version 0.3 exposed universal feasible refusal by the hybrid; version 0.4 improved the constrained bridge but failed its per-model recall floor. The frozen v0.5 comparison then produced 1,440 outputs: the repaired hybrid achieved 50.69% accuracy, 1.39% feasible recall, and zero unsafe acceptances, while the direct coordinator achieved 52.08% accuracy, complete feasible recall, and 345 unsafe acceptances. That valid negative result motivated the production semantic-admission redesign. The unexecuted study-specific v0.6 protocol is marked superseded rather than presented as evidence.

Version 0.7 demonstrates a substantial increase in hybrid utility relative to v0.5 while retaining far stronger safety than direct prompting. It nevertheless fails four of six predeclared gates, includes one unsafe ambiguity acceptance, and loses to an exact-alias control tailored to the corpus. The correct conclusion is consequently narrower than “hybrid superiority”: deterministic admission and compilation make model proposals auditable and much safer, but useful, general semantic admission remains model-sensitive and unproven.

Chapter 7

Discussion

7.1 Interpretation

The contribution is an integrative authorization/refusal boundary. Established mechanisms—typed schemas, planning constraints, access policy, task DAGs, leases, fencing, idempotency, and protocol adapters—are combined so that linguistic output cannot silently acquire executable authority. The new contribution claimed here is their operational integration and traceable refinement into one prototype, not invention of each mechanism or formal verification of arbitrary agents.

The LLM is useful where wording varies and decomposition is not known in advance. It is not an authority. Stable capability identifiers, local trust assignment, typed constraints, validation contracts, and completion contracts determine what may be dispatched. A structured rule-only oracle remains a useful upper bound because it shows what can be achieved when interpretation is already supplied; it is not a fair replacement for an end-user system receiving natural language.

Version 0.3 exposed the initial integration’s central weakness: safety was obtained through universal refusal. Version 0.4 narrowed the linguistic role and produced substantially better but model-sensitive development results. Version 0.5 then showed that the historical identifier-copying bridge did not transfer to fresh matched pairs: it retained zero unsafe acceptance only by accepting five of 360 feasible repeated observations. That negative experiment motivated the production redesign and is no longer treated as the primary comparison.

The frozen v0.7 study evaluates raw language through production semantic admission and the shared compiler. At primary seed 11, the hybrid attains 71.88% balanced accuracy and 44.79% feasible recall, with one unsafe acceptance in 192 observations. The direct LLM attains 53.13% balanced accuracy and 98.96% feasible recall but makes 89 unsafe acceptances. The pair-cluster interval for the hybrid-minus-direct unsafe-acceptance rate is entirely negative, so the safety-superiority gate passes. The recall difference is between -63.54 and -45.83 percentage points, so non-inferiority fails. Safety and utility remain in

tension rather than being jointly achieved.

Model separation is essential. Qwen3-1.7B hybrid recall is 56.25% with no unsafe acceptance; Gemma-4-E2B refuses every feasible case and therefore has zero recall; Qwen3-8B reaches 78.13% recall but produces the sole unsafe hybrid acceptance. High seed agreement—98.44% for the hybrid and 96.88% for direct—shows that the primary pattern is not explained by widespread seed instability. It does not establish backend seed compliance or generalize beyond these quantizations and this laptop runtime.

The exact-name/alias baseline solves all 64 designed cases. Its perfect balanced accuracy means the lexical-superiority gate fails decisively: the hybrid-minus-alias interval is $[-32.81, -23.44]$ percentage points. This is both a negative result for the proposal on the corpus and a construct warning. The designed language contains enough exact aliases for a deterministic matcher to recover the author labels; therefore the corpus does not establish that an LLM is necessary. Future evaluation must include independently sourced utterances whose semantics cannot be recovered through catalog phrase overlap while keeping the same safety boundary.

The production redesign nevertheless isolates where failure occurs. The oracle/compiler invariants all pass, provider selection is registry-order independent, alternative recovery succeeds, and search beyond 4,096 assignments refuses reproducibly. Eight observed primary-seed hybrid acceptances replay through the fixture A2A execution boundary with dependency artifacts and validation contracts, all completing. These results support the implemented deterministic boundary and execution path, not comparative superiority of semantic admission.

7.2 Guarantees and Non-Guarantees

Within the tested implementation, authorization precedes dispatch, unresolved constraints fail closed, declared failed dependencies block descendants, independent branches may continue, session traces are isolated, and stale fenced writes are rejected. These are implementation properties supported by tests and measured traces. They are not mathematical proofs over every deployment, scheduler, transport, or database failure.

Contract validity is weaker than semantic truth. JSON Schema can establish nested type and required-field conformance; a validator can establish a domain-specific property only if that property is encoded. Neither implies that an open-ended summary is complete or that a physical action is safe. The trusted registry and honest-agent assumptions remain explicit.

The admission boundary controls allowed origins, HTTPS policy, locally assigned trust, required credential schemes, and request-header propagation. It does not provide cryptographic identity, attestation, or defense against a malicious admitted agent. Plain HTTP is enabled only for explicit development fixtures.

The legacy PostgreSQL lease-and-fencing mode is not consensus because one database

remains its serialization point. The new etcd mode is consensus-backed: Raft orders registry, ownership, plan, attempt, and terminal mutations while a voter majority is available. This does not imply Byzantine tolerance or exactly-once external effects. Fencing prevents an obsolete coordinator from committing authoritative state only when every mutation checks the fence, and external safety additionally requires the receiving agent to enforce the highest fence and durably deduplicate operation keys.

7.3 Critical Assessment of the Distributed Implementation

The strongest design decision is the separation between logical authority and physical availability. The theory still has one accountable authorization role, while etcd removes the single volatile coordinator and single PostgreSQL coordination database from the authoritative path. Transactional owner comparisons, monotonic revision fences, attempt-before-result checks, terminal uniqueness, lease-backed agent registration, and fail-closed quorum errors reinforce one another instead of treating leader election alone as sufficient. Learner-first admission and serialized membership changes reduce reconfiguration risk. Keeping JSONL and PostgreSQL adapters also permits contract comparison and incremental migration rather than an irreversible cutover.

The unified container is operationally simpler, but it creates correlated resource and failure domains. A CPU-intensive planner, slow disk, container restart, host loss, or network namespace fault can affect both the coordinator and its local etcd member. The dispatch concurrency limit mitigates but does not eliminate heartbeat starvation. A production topology must place voters on independent hosts or failure domains, reserve CPU and durable storage for etcd, and monitor disk latency. Three containers on one Docker host demonstrate process-level behavior, not host-level fault tolerance.

Automatic scaling is also easy to overstate. The implementation automatically discovers an existing cluster and reconciles the membership of coordinator nodes that already exist. It does not create compute instances, repair a permanently lost majority, or safely choose a new cluster after loss of quorum. DNS/direct seeds remain necessary across routed networks; multicast is a link-local fallback and may be filtered by container, cloud, or enterprise networks. Target one has no fault tolerance. Targets five and seven tolerate more failures but add write latency and have not yet been exercised by the accepted harness.

The security boundary is appropriate for a trusted private prototype but weak for hostile deployment. HMAC authenticates possession of a cluster-wide secret, not an individual node identity; compromise of one coordinator therefore permits forged cluster messages until the shared secret is rotated. Nonce and timestamp checks depend on bounded clock skew and local replay caches. Discovery authentication does not encrypt or authenticate etcd peer and client transport. Deferring mTLS, certificate rotation, per-node authorization, and attestation

leaves etcd ports and shared secrets as high-value assets. The separate agent-registration secret is useful compartmentalization, but admitted malicious agents and compromised voters remain out of scope.

The custom HTTP-gateway client keeps dependencies small and supports endpoint rotation, transactions, leases, and watch resumption, but it also places subtle retry, compaction, cancellation, and error-classification behavior in project code. Quorum-unavailable errors must never be converted into permissive retries or stale reads. The asynchronous PostgreSQL projector correctly avoids a readiness dependency, yet event deletion must remain disabled until projection or export is acknowledged and a retention policy is explicit.

Consensus-v4 shows a materially stronger but still bounded distributed result. Formation at three, five, and seven voters; leader loss; minority partition; majority loss and restoration; concurrent ownership; audit failure; failed-voter replacement; all three crash windows; and the supplementary combined-fault scenario passed in every completed repetition. No executed primary safety invariant failed. Nevertheless, only 140 of 150 primary checks executed because one repetition of each live-reconfiguration path timed out before its five checks ran. These are infrastructure-indeterminate outcomes, not demonstrated safety violations. The campaign also records a dirty source tree and is therefore ineligible as accepted clean evidence. RQ5 is partially supported for the repeatedly completed conditions, while complete reconfiguration reliability and clean-source reproducibility remain open.

7.4 Claim and Terminology Discipline

The writing must distinguish three concepts that earlier drafts conflated. *Centralized authorization* describes one logical decision role; it does not necessarily mean one process. *Consensus-backed replication* describes how that role’s state is ordered and recovered; it does not make authorization a peer negotiation. *Automatic membership reconciliation* describes adding or removing already running nodes; it is not automatic compute provisioning. Accordingly, “fault tolerant” should always be qualified as crash-fault tolerant while a majority survives, and “no single point of failure” should be limited to the authoritative coordination path under independent voter failure domains. The implementation claims duplicate suppression for measured cases, not exactly-once effects.

7.5 Threats to Validity

Construct validity is protected by separating semantic fields, feasibility, execution, artifacts, trace, and recovery measures, but the author-designed and author-labelled corpus creates construct circularity and researcher-bias risk. Freezing labels before confirmatory collection and hiding them during inference reduces leakage, not bias or lack of independent adjudica-

tion. Internal validity remains threatened by model nondeterminism, unverifiable backend seed compliance, prompt sensitivity, baseline choices, fixture ownership, one separately classified model-runtime failure, and distributed checks authored with the implementation. External validity is limited to 64 held-out designed cases, three small local models, controlled fixture agents, selected PostgreSQL crash faults, and single-host Docker campaigns. Conclusion validity uses 32 matched pairs, model/category reporting, one primary seed, one stability replication, and pair-clustered bootstrap intervals. Repeated measurements cannot compensate for the small designed corpus, author labels, correlated requested-seed outputs, infrastructure-failed trials, or the absence of independent failure domains.

7.6 Defense Readiness

The repository is technically evidence-complete for the v0.7 rebuild: the complete test suite, Ruff, mypy, strengthened production branch-coverage gates, deterministic compiler benchmark, eight fixture-A2A replays, and the full three-model matrix pass their integrity checks. The 45-trial consensus-v4 matrix is complete but not accepted clean evidence because its provenance records the preserved dirty worktree. Defense readiness depends on presenting mixed results without dilution: RQ4 passes only safety superiority and symbolic invariants, while RQ5 is partially supported with two reconfiguration repetitions indeterminate. Independent corpus validation remains a disclosed limitation. Administrative tutor, tribunal, clean-source rerun, and packaging metadata remain release blockers. Independent label review and a broader low-overlap raw-language corpus are required before claiming a neutral benchmark, a superior planning pipeline, or a complete map of supported fault conditions.

Version 0.8 addresses the specific v0.7 semantic failure modes through evidence grounding, deterministic ambiguity, quotation and negation boundaries, non-authoritative retrieval, and constraint-directed provider search. Its complete development matrix passes the qualification gates for both permitted models, but this is intentionally a stopping point rather than a success claim: the same cases informed implementation revision. Qwen3-8B is excluded only from v0.8 and its v0.7 result remains unchanged. A cross-version comparison will be descriptive even after confirmation because the corpus and semantic interface changed. Until the two human labeling passes are completed and the 768 held-out observations are validated, v0.8 cannot alter the thesis’s confirmatory conclusions.

Chapter 8

Conclusions and Future Work

The thesis proposes a protocol-adaptable boundary between linguistic semantic selection and executable multi-agent coordination. RQ1 is answered at the design and prototype level: canonical capabilities, typed constraints, task dependencies, authority, validation and completion contracts, artifacts, attempts, and persisted trace identity must be explicit to justify dispatch. RQ2 is answered architecturally: language selects admitted goal, policy, contract, exclusion, and ambiguity identifiers, while deterministic components own dependency closure, typed-request construction, provider assignment, enforceable predicates, authorization, refusal, and recording. RQ3 is answered through bounded task-local auxiliary specifications whose method, schema, validator, read-only authority, and lifecycle are explicit.

RQ4 is answered by the complete frozen v0.7 matrix. At primary seed 11, the hybrid achieves 71.88% balanced accuracy, 44.79% feasible recall, and one unsafe acceptance; the direct LLM achieves 53.13% balanced accuracy, 98.96% feasible recall, and 89 unsafe acceptances. Safety superiority is supported by the paired interval, but zero unsafe acceptance, useful coordination, recall non-inferiority, and lexical superiority all fail. The exact-name/alias baseline solves all 64 designed cases, so the corpus does not establish a need for an LLM and exposes a strong construct-validity limitation. Only two of six predeclared gates pass: safety superiority and the deterministic symbolic invariants. No independent adjudication or broad superiority is claimed.

RQ5 is partially supported by the complete consensus-v4 campaign. The implementation supplies consensus-backed storage, generation-CAS configuration, serialized resumable membership operations, fencing, linearizable authorization reads, receiver-side operation controls, and fail-closed quorum responses. Forty-three of 45 trials passed and no executed primary safety invariant failed; 140 of 150 primary checks executed. However, one repetition of each reconfiguration path ended in an infrastructure timeout before its checks ran. Moreover, the preserved dirty worktree makes `evidence_valid=false`, so the campaign is descriptive complete evidence rather than accepted clean release evidence. No claim is made for multi-host availability, Byzantine tolerance, automatic recovery after majority loss, or

exactly-once external effects.

The principal scientific claim is therefore deliberately narrow. The implementation demonstrates how established symbolic, security, persistence, interoperability, and crash-consensus mechanisms can be integrated so that unsupported linguistic proposals are refused before action and authoritative mutations stop when quorum is unavailable. Duplicate external effects are suppressed only when receivers enforce both operation keys and monotonic fences. The work does not prove semantic truth, production safety, malicious-agent resistance, decentralized authorization, Byzantine tolerance, exactly-once effects, or universal protocol interoperability.

The technical evidence includes unsuccessful outcomes rather than selected reruns. It does not establish that an LLM is necessary: the exact-alias baseline solves the full v0.7 corpus, while Gemma’s production hybrid refuses every feasible primary-seed case. The deterministic compiler nevertheless passes all oracle, registry-permutation, alternative-provider, and bounded-search invariants, and eight observed hybrid acceptances complete fixture A2A replay. Remaining release work is administrative and provenance-related: supply tutor and tribunal metadata, create a clean packaging commit after author review, rerun consensus-v4 from that clean source, rebuild the signed-off PDF, tag, push, and archive the submission package. Longer-term research should independently source and adjudicate a broader raw-language corpus with low lexical overlap, improve useful semantic admission without weakening fail-closed behavior, evaluate larger provider-search spaces, investigate the two reconfiguration timeouts, introduce mTLS with per-node identities and rotation, test external domain agents, add semantic validators and replanning, place voters in independent failure domains, and evaluate snapshot and disaster-recovery procedures in broader deployments.

Version 0.8 implements the immediate semantic and provider-search improvements and completes a valid 576-observation development qualification matrix using only Qwen3-1.7B and Gemma-4-E2B. Both models pass all development gates; Qwen3-8B is explicitly excluded from v0.8 while its historical v0.7 evidence remains unchanged. These tuned outcomes are not confirmatory. The remaining scientific sequence is to complete the two output-blind human labeling passes, freeze the resulting hashes, collect all 768 held-out observations without repair or selective retry, generate evidence-bound tables and figures, and then reassess the claims while retaining any negative or mixed outcome.

Appendix A

Reproduction and Artifact Manifest

The repository root contains REPRODUCING.md, evidence-manifest.json, the locked dependency graph, Docker configurations, and the versioned corpus. Legacy reports are explicitly excluded. A final accepted manifest must contain the clean Git commit, Docker digests, corpus hash, run identifiers, and hashes of generated thesis tables.

A.1 Command Sequence

```
uv sync --locked --group dev
uv run ruff check src tests scripts
uv run mypy src/unified_multi_agent_coordination
uv run pytest --cov --cov-branch --cov-report=json
uv run python scripts/check_coverage.py coverage.json
uv run unified-coordination-scenarios --no-output
$env:EVIDENCE_RUN_DIR="demo_runs/system/<new-run-id>"
docker compose -f docker-compose.system.yml up --build \
  --abort-on-container-exit --exit-code-from system-tests
$env:EVIDENCE_RUN_DIR="demo_runs/postgres/<new-run-id>"
docker compose -f docker-compose.distributed.yml up --build \
  --abort-on-container-exit --exit-code-from distributed-system-tests
uv run unified-consensus-matrix \
  --output-dir demo_runs/consensus/<new-run-id> --trials 3 \
  --promotion-candidate
uv run unified-generate-defense-corpus-v07 --output corpus/v0.7
uv run unified-symbolic-benchmark-v07 \
  --output demo_runs/v0.7/deterministic-benchmark.json
uv run unified-defense-study-v07 --corpus corpus/v0.7 \
  --freeze-protocol
uv run unified-defense-study-v07 --corpus corpus/v0.7 --check
```

```

uv run unified-defense-study-v07 --corpus corpus/v0.7 \
  --phase confirmatory --model qwen/qwen3-1.7b
uv run unified-defense-study-v07 --corpus corpus/v0.7 \
  --phase confirmatory --model google/gemma-4-e2b \
  --resume demo_runs/v0.7/<run-id>
uv run unified-defense-study-v07 --corpus corpus/v0.7 \
  --phase confirmatory --model qwen/qwen3-8b \
  --resume demo_runs/v0.7/<run-id>
uv run unified-analyze-defense-study-v07 \
  --run demo_runs/v0.7/<run-id> --corpus corpus/v0.7 \
  --phase confirmatory \
  --benchmark demo_runs/v0.7/deterministic-benchmark.json
uv run unified-generate-defense-corpus-v08 --output corpus/v0.8
uv run unified-generate-defense-corpus-v08 \
  --output corpus/v0.8 --prepare-review
uv run unified-symbolic-benchmark-v08 \
  --output demo_runs/v0.8/deterministic-benchmark.json
uv run unified-defense-study-v08 --corpus corpus/v0.8 \
  --phase development --model qwen/qwen3-1.7b
uv run unified-defense-study-v08 --corpus corpus/v0.8 \
  --phase development --model google/gemma-4-e2b \
  --resume demo_runs/v0.8/<development-run-id>
uv run unified-analyze-defense-study-v08 \
  --run demo_runs/v0.8/<development-run-id> \
  --corpus corpus/v0.8 --phase development
# Complete and reconcile both output-blind author review worksheets.
uv run unified-defense-study-v08 --corpus corpus/v0.8 \
  --freeze-protocol
uv run unified-defense-study-v08 --corpus corpus/v0.8 \
  --phase confirmatory --model qwen/qwen3-1.7b
uv run unified-defense-study-v08 --corpus corpus/v0.8 \
  --phase confirmatory --model google/gemma-4-e2b \
  --resume demo_runs/v0.8/<confirmatory-run-id>
uv run unified-analyze-defense-study-v08 \
  --run demo_runs/v0.8/<confirmatory-run-id> \
  --corpus corpus/v0.8 --phase confirmatory
uv run unified-defense-study --check
uv run unified-analyze-defense-study \
  demo_runs/v0.3/20260713T013624Z-b9fd8b6f83 \
  --corpus corpus/v0.3 \
  --output demo_runs/v0.3/20260713T013624Z-b9fd8b6f83/analysis.json

```

```
uv run unified-defense-study-v04 --corpus corpus/v0.4 --check
uv run unified-analyze-defense-study-v04 \
  --run demo_runs/v0.4/20260713T121506Z-a0ecd2bf0e \
  --corpus corpus/v0.4 \
  --output demo_runs/v0.4/20260713T121506Z-a0ecd2bf0e/<new-analysis>.json
uv run unified-runtime-ablation-study \
  --output-dir demo_runs/runtime_ablation/<new-run-id>
uv run unified-evidence-preflight --require-release \
  --corpus corpus/v0.5 --repository .
```

Appendix B

Predicate Refinement and Claim–Evidence Matrix

Claim or predicate	Implementation refinement	Evidence required
Coverage	Canonical capability ID, full modes and schema coverage	Feasibility and ID-perturbation tests
Authority	Local trust order and admission policy	Remote-card and credential tests
Constraints	Typed source, JSON pointer, operator and expected value	Missing, type, false and unresolved cases
Verifiability	Task, requirement or capability validation contract	Validatorless and nested-schema tests
Dependency safety	Completed prerequisites and declared artifacts only	Failure/timeout/unknown branch tests
Traceability	Persisted session, plan, task and attempt events	Ordering, restart and isolation tests
Recovery	Lease, fencing, idempotency and unknown-side-effect policy	PostgreSQL crash-fault harness
Distributed authority	etcd/Raft state, learner-first membership, generation CAS, resumable phases, and quorum fail-closed behavior	Historical three-voter report plus required clean 3/5/7 campaign
RQ4 comparison	Frozen production-path, author-labelled, label-hidden v0.7 matched-pair study	Exact 768-output dual-arm matrix, per-model analysis, deterministic controls, clustered intervals, and reproducible digest

Runtime mechanisms	Secure-default policy surfaces and explicit negative controls	Separate 350-observation runtime-only bundle
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B.1 Research-Question Traceability

RQ	Claim	Mechanism and evidence	Current boundary
RQ1	Executability requires explicit identity, coverage, authority, dependencies, artifacts, validation, and completion.	Typed models, feasibility predicates, hydration, artifact and trace tests.	Prototype predicates; no proof for arbitrary domains.
RQ2	Linguistic proposal can be separated from deterministic authorization.	Non-authoritative drafts, symbolic gate, authorization-before-dispatch tests, and four v0.5 analysis views.	Implemented separation; author-labelled cases and small local models.
RQ3	Temporary language capabilities can be bounded and auditable.	Task-local auxiliary specification, admission and expiry tests, separate runtime ablation.	Read-only fixture scope; no hostile generated code.
RQ4	The constrained bridge is compared fairly with a direct LLM coordinator.	Complete 768-output v0.7 matrix, equal public inputs and call budgets, exact-alias and oracle controls, seed stability, and pair-cluster bootstrap.	Conclusions are limited to the observed frozen gates and author-only labels.

RQ5	A replicated coordinator should preserve one authorization state under measured crash and reconfiguration faults while a majority survives.	etcd store, monotonic fences, durable serialized membership state, quorum error contract, and complete 45-trial consensus-v4 campaign.	Single-host crash-fault evidence; no Byzantine or independent-failure-domain claim.
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Appendix C

Tribunal Questions and Defensible Boundaries

What is novel? The protocol-adaptable integration and operational refinement of a linguistic proposal boundary, symbolic authorization/refusal, durable evidence, admission policy, and measured recovery. Individual mechanisms are prior art.

Why use an LLM? Natural-language interpretation and candidate decomposition vary semantically; a structured rule oracle assumes that work is already done. The LLM remains non-authoritative.

What is guaranteed? The tested implementation enforces declared predicates and ordering under its trusted-code and trusted-store assumptions. It does not guarantee semantic truth or runtime success.

What is trusted? The coordinator code, durable store, locally configured trust policy, admitted registry descriptions, and contract validators. Malicious admitted agents remain outside scope.

Is fencing consensus? Fencing alone is not consensus. The legacy PostgreSQL mode serializes through one database; the etcd mode separately uses Raft consensus to order authoritative state. Fencing then prevents stale session owners from committing and must also be enforced by effectful receivers.

Does the cluster scale automatically? Membership of already running coordinator nodes is discovered and reconciled automatically. The software does not provision compute, and permanent majority loss requires operator-led disaster recovery.

Is there no single point of failure? With at least three voters placed in independent failure domains, no single coordinator or etcd-member crash stops the authoritative path. One

voter, one Docker host, shared network infrastructure, and a cluster-wide secret remain single-failure or correlated-failure risks.

How broad is A2A interoperability? Controlled fixtures plus one vendored, upstream-derived sample pinned to an upstream commit. This is evidence of concrete compatibility, not universal conformance.

Did the hybrid work empirically? Only as a fail-closed boundary. It made zero unsafe acceptances but accepted five of 360 feasible repeated observations after repair. It was neither useful nor more accurate than the direct arm on this corpus.

Why trust the comparison? The corpus and author labels were frozen before collection, labels were hidden until both arms completed, all 1,440 identities and equal call budgets were verified, and independent analysis runs reproduced the same digest. Labels were not independently adjudicated, so claims remain limited to framework conformance on this designed corpus.

Is RQ5 answered? Partially. Consensus-v4 completed 45 trials: 43 passed, two reconfiguration trials had infrastructure timeouts, and no executed primary invariant failed. Because the preserved worktree was dirty, the campaign is not accepted clean release evidence; a clean-source rerun is still required.

Appendix D

Disclosure of Artificial Intelligence Use

Artificial-intelligence systems were used in two distinct roles during this thesis: as experimental subjects whose outputs form part of the evaluation, and as assistive tools used during research, software development, and manuscript revision. These roles must not be conflated. Experimental model outputs are immutable study observations; assistive outputs are suggestions that remain subject to author review. In every role, responsibility for the research design, reference labels, source selection, implementation, interpretation, claims, and submitted text remains with the author.

D.1 Systems, Tools, and Purposes

System or tool	Role	Use in the thesis
OpenAI Codex (GPT-5-based project sessions)	Interactive research, writing, and software-development assistant	Used to critique chapter structure and claims; assist with literature discovery and source triage; draft or revise prose, documentation, and the bilingual abstract; propose code, refactorings, tests, evaluation scripts, CI checks, and reproducibility instructions; inspect claim–evidence consistency; and assist with LaTeX/PDF quality review. Suggested citations were not accepted as evidence without inspection of the underlying source. All changes remained visible in the repository and were reviewed against tests, static checks, evidence files, or the compiled manuscript.
LM Studio	Local model-serving environment	Used to load and serve the pinned local models through an OpenAI-compatible HTTP interface for the v0.3–v0.5 historical protocols and the v0.7 Qwen-first production-path protocol. It provided the execution environment for study inference; it was not treated as an author, adjudicator, or source of scientific claims.

System or tool	Role	Use in the thesis
Qwen3 1.7B, Gemma 4 E2B, and Qwen3 8B	Experimental language models	Used as study subjects to produce non-authoritative semantic selections or direct coordination decisions under frozen prompts, schemas, cases, arms, and requested seeds. The v0.7 confirmatory protocol preserves each raw response and treats invalid schema as failure without semantic repair. These models did not receive hidden labels and were not used as independent validators of those labels.
Lingo	Typed linguistic-application framework	Used inside the prototype to expose bounded interpretation, extraction, classification, and candidate-planning capabilities through explicit state and schemas. Lingo is implementation middleware rather than the source of the evaluated model weights or an authority over coordination decisions.
OpenAI- compatible client interfaces	Model-transport interface	Used to send local requests to LM Studio and record responses, metadata, latency, and errors. Use of the interface does not imply that hosted OpenAI models generated the experimental observations.

AI assistance also contributed to exploratory scenario, prompt, baseline, and test-design critique. The author selected the final protocol, determined and froze the reference labels, and accepted responsibility for their correctness. AI assistance is therefore not presented as independent annotation or adjudication; the absence of an independent human adjudicator remains a threat to validity.

D.2 Authorial Control and Verification

AI-generated or AI-revised material was not accepted solely because it was fluent or technically plausible. Manuscript changes were checked against primary sources and repository

evidence. Code changes were inspected and exercised through the applicable Ruff, mypy, pytest, coverage, deterministic, container, and evidence-preflight checks. Numerical statements were derived from immutable JSON evidence or generated LaTeX macros. Statistical analyses were implemented as deterministic programs and reproduced by digest. Figures in the submitted manuscript are repository-authored LaTeX/TikZ artifacts; no generative image system was used for the current figures.

The AI assistants did not determine the final research questions, authorize execution in the evaluated system, alter collected model outputs, fabricate measurements, select favorable runs, assign themselves authorship, or replace the author's interpretation of negative outcomes. Working conversations with an assistant may contain provisional ideas and rejected text; they are not research evidence. The version-controlled manuscript, source code, frozen protocols, raw outputs, analysis programs, manifests, and accepted reports are the auditable artifacts.

For clarity, the principal non-AI tools used to construct and validate the artifact include Python and uv; Pydantic, FastAPI, HTTPX, and Uvicorn; pytest, coverage.py, Ruff, and mypy; Git and GitHub Actions; Docker Compose; PostgreSQL; etcd/Raft; the A2A SDK and upstream-derived sample; JSON Schema; and LuaLaTeX, BibTeX, TikZ, Poppler, and PDF inspection tools. These tools execute deterministic software, infrastructure, validation, or document-processing functions and are not reported as generative-AI authorship assistance.

No other generative-AI assistant is documented in the project record. If an additional assistant was used outside that record, it must be added to this disclosure before submission. This statement is intended to make the role of AI inspectable without transferring scholarly responsibility away from the author.

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